



Rough set theory applied to finite dimensional vector spaces

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ABSTRACT

In this paper, we study finite dimensional vector spaces using rough set theory (RST) by defining a Boolean information system I_B associated with a vector space V for a given basis B . We define an indiscernibility relation on V and investigate different partitions induced on V . We identify that every reduct of the information system is a basis of V and the core is empty in the case of V over field F with $|F| \geq 3$. We study the measure of dependency between different subsets of V . Moreover, we define three posets and prove that these posets are isomorphic and complete. We study lower and upper approximations for different subsets of V and determine rough and exact sets. We give the general form of the entries of the discernibility matrix. We determine essential sets and the essential dimension of the information system, and prove that minimal entries of the discernibility matrix coincide with the essential sets. Finally, we demonstrate the use of the developed theory by providing two practical examples.

1. Introduction

In data science, pattern recognition and machine learning, vector spaces are fundamental for representing data instances. Each data instance corresponds to a vector, with components corresponding to specific feature value where a feature is a measurable characteristic of objects under study. Features can be numerical or categorical. Many machine learning algorithms can handle only numerical features, so categorical features are converted into numerical features to make them suitable for implementation of algorithms. Vector spaces and associated concepts provide terminology and framework in various tasks like identification of similarities, classification, regression, and clustering. These spaces, also called feature spaces, and the vectors, also called feature vectors, are crucial for data preprocessing, model building, and feature engineering. Feature spaces are fundamental tool for representing data in machine learning and many other data mining problems. Feature spaces have been used in quantum machine learning, knowledge recommendation systems, clustering numerical data and deep learning systems [4,13,24,36].

Real-world data sets often have many redundancies. Reducing these redundancies in data reduce complexity of data and makes the data analysis efficient. The traditional representation of data is in the form of data tables consisting of object-attribute information such that each row represents an object and each column represents an attribute or a feature. Each row in this data table is an n -dimensional feature vector, where n is the number of features. This representation of data provides necessary framework to study data using RST. The set of features or attributes yield an indiscernibility relation (equivalence relation) on the objects, which is the main building block of RST. The indiscernibility relation performs granulation on the set on which it is defined and induces a partition on it, where each block in the partition is called a granule. These granules are then used to find lower and upper approximations of sets. In RST, a subset of a universal set is characterised by two crisp sets known as the lower and upper approximations of the set.

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RST has been proven one of the efficient mathematical techniques for analysing information systems and extracting insights from them [18,20,27,29]. RST has been applied to classification problems, decision analysis, pattern recognition, knowledge discovery, information systems, intelligent systems, and many other fields [5,15,21,25–28].

The key concept in RST is indiscernibility relation which inspires the study of mathematical structures using terminology and concepts of RST. Kuroki and Wang [17] gave the concept of a rough subgroup with respect to a normal subgroup of a group and defined a type of rough subgroup of a fuzzy normal subgroup. In [16], Kuroki introduced the concept of a rough ideal in a semigroup, whereas Davvaz in [10] gave the idea of rough subrings with respect to an ideal of a ring. In [22], the notions of rough groups and rough subgroups were introduced. Estaji and Hooshmandasl [12] studied lattice theory in the notion of RST and defined lower and upper rough ideals in a lattice. Järvinen [14] introduced some foundations of rough sets in the context of lattice theory. Wei and Qi [31] identified the relationship between lattice reduction and rough set reduction. Zhu and Wang [35] introduced the concept of rough matroids. Moreover, in [23] Milan identified that some well established results in algebraic and topological notions also exist in RST.

Vector space is a well known algebraic structure, similar to other algebraic structures, vector spaces have been studied in notion of RST in different perspectives. To the best of our knowledge, Wu, Xie and Cao [33] initiated the study of vector spaces in context of RST. The authors studied approximations of different subsets of vector space with respect to subspaces of the vector space using congruence relation as an indiscernibility relation. Wu and Xie [32] studied roughness in vector spaces and introduced lower and upper rough subspaces. Almohammadi and Ozel [2] proposed the idea of rough vector spaces, studied its algebraic structure and introduced topological rough vector space. Moreover, roughness in hypervector spaces and approximations in hypervector spaces were studied in [1].

In this paper, we study vector spaces using the terminology of RST by defining a Boolean information system with it. We define a Boolean information system for a given $\mathcal{V}$ with respect to a basis B of $\mathcal{V}$ and define an indiscernibility relation on $\mathcal{V}$. Using the defined indiscernibility relation as a key concept, we study different partitions induced by different subsets of a vector space and investigate relationships between these partitions. We find reducts of information systems and observe that reducts depend on the underlying basis set of the information system and the core of the information systems associated with vector spaces over a field having cardinality three or more is empty. Moreover, we define three types of posets with vector spaces and observe that these posets are complete lattices only for one and two-dimensional vector spaces. We study approximations of different subsets of a vector space and identify rough and exact sets. Furthermore, we define the discernibility matrix and numerical discernibility matrix associated with the information table. We define a poset with distinct entries of the discernibility matrix and prove that the posets form complete lattices only for the vector spaces over the field of cardinality two. We define essential sets and provide a basic link between the essential sets and entries of the discernibility matrix of the information system. Moreover, we provide two real-life examples to demonstrate use of the theory developed. Throughout this paper, we only consider vector spaces of finite dimension. The main contribution of this paper is that we provide a novel method of studying vector spaces (feature spaces) using the notions of RST which will aid in many data mining and machine learning problems. For the purpose of this paper, the terms vector space and feature space are used interchangeably.

The contents of the paper are organized as follows. In section 2, we provide elementary terminology and concepts needed for the paper. In section 3, we define a Boolean information system for a vector space in the context of a basis of the vector space. We define an indiscernibility relation that originated from the defined information system and study different partitions of the vector space, and reducts of the information system. In section 4, we study the dependency measure of different subsets of a vector space and associated three types of posets with vector spaces and discuss their completeness. In section 5, we study the lower and upper approximation of different subsets of a vector space. Moreover, we defined approximation accuracy and rough membership function and studied their properties. In section 6, we define the discernibility matrix and the numerical discernibility matrix and determine the essential sets of the information systems. In section 7, we examine two real-life problems to illustrate the application of the proposed concepts. Finally, we provide the conclusion of this paper in the last section.

2. Basic concepts and terminology

In this section, we recall some elementary concepts of RST and provide other terminology used in the sequel; for details of these concepts, we refer the reader to [9,19,25–27,30].

2.1. Rough set theory

An indiscernibility relation which is a binary relation that satisfies all the axioms of an equivalence relation is the main building block of RST. The indiscernibility relation partitions the set on which it is defined and forms indiscernibility blocks.

Let $\mathfrak{R}$ be an indiscernibility relation on a set $\mathcal{Z}$. The indiscernibility relation partitions $\mathcal{Z}$ into disjoint sets, such that the intersection of any two sets in the partition has an empty intersection whereas the union of all the sets in a partition is equal to $\mathcal{Z}$. Each set in the partition is an indiscernibility block and all the elements in an indiscernibility block are said to be indiscernible with respect to $\mathfrak{R}$. We denote by $[z]_{\mathfrak{R}}$, the indiscernibility block containing $z \in \mathcal{Z}$.

Definition 2.1. [25] For $Y \subseteq \mathcal{Z}$, the lower and upper approximation of Y is defined respectively as

$$L_{\mathfrak{R}}(Y) = \{z \in \mathcal{Z} : [z]_{\mathfrak{R}} \subseteq Y\},$$

$$U_{\mathfrak{R}}(Y) = \{z \in \mathcal{Z} : [z]_{\mathfrak{R}} \cap Y \neq \emptyset\},$$

If $L_{\mathfrak{R}}(Y) = U_{\mathfrak{R}}(Y)$ then Y is called exact; otherwise, it is called rough. Moreover, it is proved in [25] that for any $Y \subseteq \mathcal{Z}$, $L_{\mathfrak{R}}(Y) \subseteq Y \subseteq U_{\mathfrak{R}}(Y)$. Note that if Y is exact, then $L_{\mathfrak{R}}(Y) = Y$. The approximation accuracy of Y is defined by $\alpha_{\mathfrak{R}}(Y) = |L_{\mathfrak{R}}(Y)| / |U_{\mathfrak{R}}(Y)|$. Following are some properties of lower and upper approximations in terms of approximation accuracy.

1. If $L_{\mathfrak{R}}(Y) = U_{\mathfrak{R}}(Y)$ then $\alpha_{\mathfrak{R}}(Y) = 1$.
2. If $L_{\mathfrak{R}}(Y) = \emptyset$ then $\alpha_{\mathfrak{R}}(Y) = 0$.
3. If $\emptyset \neq L_{\mathfrak{R}}(Y) \subset Y$ then $0 < \alpha_{\mathfrak{R}}(Y) < 1$.

Let $\mathfrak{R}_1$ and $\mathfrak{R}_2$ be two indiscernibility relations defined on $\mathcal{Z}$ and $\Pi_{\mathfrak{R}_1}, \Pi_{\mathfrak{R}_2}$ be the partitions induced by $\mathfrak{R}_1$ and $\mathfrak{R}_2$ on $\mathcal{Z}$, respectively. The dependency of $\mathfrak{R}_1$ on $\mathfrak{R}_2$ is defined by $\Upsilon_{\mathfrak{R}_1}(\mathfrak{R}_2) = |POS_{\mathfrak{R}_1}(\mathfrak{R}_2)| / |\mathcal{Z}|$, where $POS_{\mathfrak{R}_1}(\mathfrak{R}_2) = \{z \in \mathcal{Z} : [z]_{\mathfrak{R}_1} \subseteq [z]_{\mathfrak{R}_2}\}$ is the positive region of $\mathfrak{R}_2$ with respect to $\mathfrak{R}_1$. Note that the value of the dependency measure lies in $[0, 1]$.

2.2. Information system and indiscernibility relation

The notions of RST have been applied to information systems for identifying similarities among objects, hidden patterns in the data, and extracting important set of features.

A tuple $I = \langle U, \mathbb{A}, V_a, f \rangle$ is called an information system, where U is a set of objects, $\mathbb{A}$ is a set of features, V_a is a set of feature values and f is an information map defined as $f : U \times \mathbb{A} \rightarrow V_a$.

Note that corresponding to each object $x \in U$, there is a feature vector $(f(x, a_1), f(x, a_2), \dots, f(x, a_n))$ where $\mathbb{A} = \{a_1, a_2, \dots, a_n\}$ and $f(x, a_j)$ is the feature value of object x corresponding to feature a_j .

In [26], it is shown that every $\mathcal{A} \subseteq \mathbb{A}$ corresponds to an indiscernibility relation on U .

Definition 2.2. [26] For $\mathcal{A} \subseteq \mathbb{A}$, $Ind(\mathcal{A}) = \{(x, y) \in U \times U : f(x, a) = f(y, a), \forall a \in \mathcal{A}\}$ is the $\mathcal{A}$ -indiscernibility relation.

Since each feature subset $\mathcal{A} \subseteq \mathbb{A}$ corresponds to an indiscernibility relation and therefore induce a partition $\Pi_{\mathcal{A}}$ on U . Two feature subsets $\mathcal{A}_1, \mathcal{A}_2 \subseteq \mathbb{A}$ are indiscernible if $\Pi_{\mathcal{A}_1} = \Pi_{\mathcal{A}_2}$, and $[\mathcal{A}]_{\sim}$ denotes the collection of those subsets of $\mathbb{A}$ which are indiscernible to $\mathcal{A}$ [6].

RST has been proven an efficient tool for extracting important set of features known as the core of the information system, where the core of an information system is the intersection of all reducts of I . A minimal feature subset $\mathcal{A}$ is called a reduct of I , if the partition induced by $\mathcal{A}$ on U is the same as $\Pi_{\mathbb{A}}$. Note that an information system may have more than one reducts.

Definition 2.3. [26] A subset $\mathcal{A} \subseteq \mathbb{A}$ is core of I , if $\Pi_{\mathbb{A}} = \Pi_{\mathcal{A}}$, and $\Pi_{\mathbb{A}} \neq \Pi_{\mathbb{A} \setminus \{a\}}, \forall a \in \mathcal{A}$.

Note that the core consists of all indispensable features such that removal of these features results in loss of information provided by the complete feature set.

The reducts of an information system I can be determined from discernibility matrix and discernibility function of I [30]. A discernibility matrix is a symmetric square matrix of order $|U| \times |U|$.

Definition 2.4. [30] For I , the entry in the discernibility matrix corresponding to $x, y \in U$ is defined as $m(x, y) = \{a \in \mathbb{A} : f(x, a) \neq f(y, a)\}$, and the discernibility function is defined as $f_{dis} = \bigwedge \{ \bigvee m(x, y) : m(x, y) \neq \emptyset, \forall x, y \in U \}$.

The expression $\bigvee$ is a disjunction operator and $\bigwedge$ is a conjunction operator. If a discernibility matrix contains singleton entries in it, then the core is the collection of all those singleton entries [26]. Note that not every information system has a core such that the intersection of all reduct sets is the empty set; in such a case, extended cores, also known as essential sets, are used.

Definition 2.5. [7] For I , a subset $\mathcal{E} \subseteq \mathbb{A}$ is I -essential if $\Pi_{\mathbb{A}} \neq \Pi_{\mathbb{A} \setminus \mathcal{E}}$, and $\Pi_{\mathbb{A}} = \Pi_{\mathbb{A} \setminus \mathcal{F}}, \forall \mathcal{F} \subset \mathcal{E}$.

2.3. Vector spaces, posets and lattices

A field F is a set, such that it is an abelian group under addition, and $F \setminus \{0\}$ is an abelian group under multiplication; these operations are connected by the distributivity of multiplication over addition axiom, called as distributive law, mathematically written as: $k \cdot (l + m) = (k \cdot l) + (k \cdot m)$, for all $k, l, m \in F$. The fields $\mathbb{R}, \mathbb{C}$ and $\mathbb{Q}$ are examples of infinite fields whereas $\mathbb{Z}_p$, where p is a prime number, is an example of finite field. The smallest field is $\mathbb{Z}_2 = \{0, 1\}$, so the minimum cardinality of a field is two.

Definition 2.6. [19] A set $\mathcal{V}$ is called a vector space over F if $\mathcal{V}$ is an abelian group under addition and satisfies the following properties.

1. If $k \in F$ and $v \in \mathcal{V}$ then $kv \in \mathcal{V}$.
2. $k(u + v) = ku + kv$ for all $k \in F$ and $u, v \in \mathcal{V}$.
3. $(k + l)u = ku + lu$ for all $k, l \in F$ and $u \in \mathcal{V}$.

4. $k(lv) = (kl)v$ for all $k, l \in F$ and $v \in \mathcal{V}$.
5. For $k = 1$, $kv = v$ for all $v \in \mathcal{V}$.

A vector $v \in \mathcal{V}$ is a linear combination of vectors $v_1, v_2, \dots, v_s \in \mathcal{V}$, if $v = \sum_{i=1}^s k_i v_i$ where $k_1, k_2, \dots, k_s \in F$. A vector written in linear combination of other vectors provides information about dependency between vectors. The concept of linear dependence among vectors in $\mathcal{V}$ refers that one vector can be generated from other vectors. In other words, we say that a vector v is generated by the combination of some vectors such that these vectors are the building blocks of the vector v . Formally, the vectors $v_1, v_2, \dots, v_s$ are called linearly dependent, if there exist $k_1, k_2, \dots, k_s \in F$ not all are zero such that $\sum_{i=1}^s k_i v_i = 0$; otherwise, the vectors are called linearly independent. Moreover, if two vectors are scalar multiples of each other then they are linearly dependent vectors. The null vector θ is linearly dependent to every vector in $\mathcal{V}$.

Definition 2.7. [19] A subset of $\mathcal{V}$ is known as a spanning set of $\mathcal{V}$, if any vector in $\mathcal{V}$ can be written in the linear combination of the vectors in it. Moreover, a subset $B \subseteq \mathcal{V}$ is called a basis of $\mathcal{V}$, if B is a set of independent vectors in $\mathcal{V}$ and B is also a spanning set of $\mathcal{V}$, and the cardinality of B is called dimension of $\mathcal{V}$.

The following example demonstrates that the RGB colorspace is a vector space of dimension three [3,11].

Example 2.1. In RGB colorspace, each color is generated by combination of three colors red(R), green(G) and blue(B). These colors are independent of each other and form a basis of all other colors because all the colors are generated by the combination of these colors. For instance, the color yellow is formed by combining red and green color and purple is formed by combining red and blue. So yellow is dependent on red and green; whereas purple is dependent on red and blue. Additionally, the color black is formed by reducing the intensity of all of the three basis colors to zero, so black is the zero vector of this colorspace.

A poset (partially ordered set) is a pair $(\mathbb{P}, \leq)$, where $\mathbb{P}$ is a set and $\leq$ is a partial order on $\mathbb{P}$. The partial order $\leq$ is a binary relation on $\mathbb{P}$ which is reflexive, anti-symmetric and transitive. For a poset $(P, \leq)$ and $a, b \in P$, if $a \leq b$, we say b covers a . Moreover, if $a \neq b$ and $a \leq b$, we write $a < b$. An element $m \in \mathbb{P}$ is called meet (infimum) of a and b , if $m \leq a$ and $m \leq b$, and an element $j \in \mathbb{P}$ is called join (supremum) of a and b , if $a \leq j$ and $b \leq j$.

A poset is called a lattice if at least one pair of elements in it has a supremum and an infimum. A poset in which every pair of elements has a supremum is called a join semi-lattice, and if every pair of elements has an infimum is called a meet semi-lattice. A poset is a complete lattice, if it is both meet and join semi-lattice. With each lattice, a Hasse diagram is associated which is a graphical (pictorial) representation of the lattice.

3. Vector space as an information system and indiscernibility relation

In this section, we provide an interpretation of a vector space $\mathcal{V}$ of finite dimension n as a Boolean information system. We define an indiscernibility relation on $\mathcal{V}$ with respect to $W \subseteq \mathcal{V}$, and investigate partitions induced on $\mathcal{V}$ by different subsets W of $\mathcal{V}$.

Basis of a vector space $\mathcal{V}$ is a set of vectors that can generate any vector of $\mathcal{V}$. The set consisting of those $b_i \in B$ which have non-zero coefficient in the linear combination of $v \in \mathcal{V}$ with respect to B is named as the skeleton of v [8]. Equivalently, the set of all those basis vectors $b_i \in B$ which are necessary to generate a vector $v \in \mathcal{V}$ is named as the skeleton of v , denoted by $S_{v|B}$. From Example 2.1, the skeleton of yellow is {red, green} whereas the skeleton of purple is {red, blue}. Mathematically, skeleton of v is defined as

$$S_{v|B} = \{b_i \in B : v = \sum_{i=1}^m k_i b_i, k_i \in F \setminus \{0\}\} \quad (1)$$

where $m \leq n$. Clearly, the skeleton of null vector $\theta = (0, 0, \dots, 0)$ is $S_\theta = \emptyset$. In the context of feature space, skeleton consists of those features which are used to describe feature vector of object under study.

Example 3.1. For the 2-dimensional vector space over $\mathbb{Z}_3$, if $B = \{(1,0), (0,1)\}$ then $S_{(2,0)|B} = \{(1,0)\}$ and $S_{(2,1)|B} = \{(1,0), (0,1)\}$.

It is interesting to note that the skeleton of two linearly dependent vectors is same, whereas the converse does not hold.

Example 3.2. For the 2-dimensional vector space over $\mathbb{Z}_3$, if $B = \{(1,0), (0,1)\}$ then $(1,1) = 1(1,0) + 1(0,1)$ and $(1,2) = 1(1,0) + 2(0,1)$. Note that the vectors $(1,1)$ and $(1,2)$ are dependent on both basis vectors $(1,0)$ and $(0,1)$ but they are not dependent on each other. So, $S_{(1,1)} = S_{(1,2)} = B$ but $(1,1)$ and $(1,2)$ are linearly independent vectors.

Now we define an information system associated to $\mathcal{V}$ generated with respect to a basis B of $\mathcal{V}$.

Definition 3.1. We call $I_B = \langle \mathcal{V}, \mathcal{V}, \{0,1\}, \mathcal{F}_B \rangle$ the information system of $\mathcal{V}$ with respect to B , where $\mathcal{V}$ is considered as the universal set as well as the feature set and the information map $\mathcal{F}_B : \mathcal{V} \times \mathcal{V} \rightarrow \{0,1\}$ is defined as

Table 1
Information system of a 2-dimensional vector space over $\mathbb{Z}_3$, where $B = \{(1,0), (1,1)\}$.

	(0,0)	(1,0)	(2,0)	(0,1)	(0,2)	(1,1)	(2,1)	(1,2)	(2,2)
(0,0)	0	0	0	0	0	0	0	0	0
(1,0)	0	1	1	1	1	0	1	1	0
(2,0)	0	1	1	1	1	0	1	1	0
(0,1)	0	1	1	1	1	1	1	1	1
(0,2)	0	1	1	1	1	1	1	1	1
(1,1)	0	0	0	1	1	1	1	1	1
(2,1)	0	1	1	1	1	1	1	1	1
(1,2)	0	1	1	1	1	1	1	1	1
(2,2)	0	0	0	1	1	1	1	1	1

$$\mathcal{F}_B(u, v) = \begin{cases} 1 & \text{if } S_{u|B} \cap S_{v|B} \neq \emptyset \\ 0 & \text{if } S_{u|B} \cap S_{v|B} = \emptyset. \end{cases} \quad (2)$$

Note that every information system yields an information table and every information table corresponds to an information system. In the sequel, the notation I_B is used for both information system and information table.

Example 3.3. For the 2-dimensional vector space over $\mathbb{Z}_3$, if $B = \{(1,0), (1,1)\}$ then $S_{(0,0)|B} = \emptyset$, $S_{(1,0)|B} = S_{(2,0)|B} = \{(1,0)\}$, $S_{(1,1)|B} = S_{(2,2)|B} = \{(1,1)\}$, $S_{(0,1)|B} = B$ because $(0,1) = 2(1,0) + (1,1) = (0,1)$, and $S_{(0,2)|B} = B$ because $(0,2) = (1,0) + 2(1,1) = (0,1)$. Similarly, $S_{(2,1)|B} = S_{(1,2)|B} = B$ because $(2,1) = (1,0) + (1,1)$ and $(1,2) = 2(1,0) + 2(1,1)$. Table 1 is the information table associated with 2-dimensional vector space over $\mathbb{Z}_3$, where $B = \{(1,0), (1,1)\}$.

Since there exist more than one basis sets of a vector space and each of these basis sets generate the same vector space, it is interesting to examine whether the information tables yielded by two distinct basis of $\mathcal{V}$ are same. From equation (1), it is clear that the skeleton of a vector depends on the underlying basis B such that a vector may have different skeletons depending on the choice of B . The fact that skeleton of a vector varies according to the choice of B implies that not all the information tables associated to a vector space are same. In the following result, we provide the condition under which two distinct basis induce same information table.

Proposition 3.1. For two distinct basis B and $\hat{B}$ of a vector space, $I_B = I_{\hat{B}}$ if for each $b \in B$ there exists $\hat{b} \in \hat{B}$ such that $b = k\hat{b}$, where $k \in F \setminus \{0\}$. Moreover, for a one-dimensional vector space $\mathcal{V}$, the information table associated with $\mathcal{V}$ for any basis is the same.

Proof. Suppose B and $\hat{B}$ are two distinct basis of $\mathcal{V}$ and for each $b \in B$ there exists $\hat{b} \in \hat{B}$ such that $b = k\hat{b}$, where $k \in F \setminus \{0\}$, which implies $S_{b|B} = S_{\hat{b}|\hat{B}}$, and $\mathcal{F}_B(u, \hat{u}) = \mathcal{F}_{\hat{B}}(u, \hat{u})$ for all $u, \hat{u} \in \mathcal{V}$. Therefore, $I_B = I_{\hat{B}}$. Now suppose $\mathcal{V}$ is a vector space of dimension one then for any basis B of $\mathcal{V}$ $S_{u|B} = S_{v|\hat{B}}$ for all $u, v \in \mathcal{V}$, which implies the information table associated with $\mathcal{V}$ for any basis is the same.

Note that in the case of finite field, the information system I_B yields an information table with $|\mathcal{V}|$ rows and $|\mathcal{V}|$ columns, whereas in the case of infinite field, the information table I_B yields an infinite table. Since two vectors having same skeleton behave alike, so each of such collection of vectors can be represented as one class. We denote such classes of vectors by $C_{u|B}$ and write $C_{u|B} = \{v \in \mathcal{V} : S_{u|B} = S_{v|B}\}$. These classes provide the foundation to represent information tables associated with infinite vector spaces over an infinite field with a finite number of rows and columns. We now define an equivalent class-based information system.

Definition 3.2. We call $\bar{I}_B = \langle \bar{\mathcal{V}}, \bar{\mathcal{V}}, \{0, 1\}, \bar{\mathcal{F}}_B \rangle$ the class-based information system, where $\bar{\mathcal{V}} = \{C_{u|B} : u \in \mathcal{V}\}$ and $\bar{\mathcal{F}}_B$ is defined as

$$\bar{\mathcal{F}}_B(C_{u|B}, C_{v|B}) = \begin{cases} 1 & \text{if } S_{u|B} \cap S_{v|B} \neq \emptyset \\ 0 & \text{if } S_{u|B} \cap S_{v|B} = \emptyset. \end{cases} \quad (3)$$

The same information system, also referred as quotient information system, can be defined for vector spaces over finite fields as well.

We refer the rows of an information system I_B as a representation of vectors in $\mathcal{V}$ with respect to W .

Definition 3.3. Let $W = (w_1, w_2, \dots, w_s)$ be an ordered subset of $\mathcal{V}$, the representation of u with respect to W is defined as

$$\Gamma_B(u, W) = (\mathcal{F}_B(u, w_1), \mathcal{F}_B(u, w_2), \dots, \mathcal{F}_B(u, w_s)). \quad (4)$$

This representation of vectors defines an indiscernibility relation on $\mathcal{V}$ with respect to $W \subseteq \mathcal{V}$, two vectors $u, v \in \mathcal{V}$ are indiscernible, denoted by $u \equiv_{W|B} v$, if $\Gamma_B(u, W) = \Gamma_B(v, W)$. Equivalently, $u \equiv_{W|B} v$ if $\mathcal{F}_B(u, w) = \mathcal{F}_B(v, w)$ for all $w \in W$. If $W = \{w\}$, we write $u \equiv_w v$. We set now the collection of all vectors which are indiscernible to vector v with respect to $W \subseteq \mathcal{V}$ as:

$$[v]_{W|B} = \{u \in \mathcal{V} : \mathcal{F}_B(v, w) = \mathcal{F}_B(u, w), \forall w \in W\}.$$

The indiscernibility relation $\equiv_{W|B}$ induces a partition on $\mathcal{V}$, denoted by $\Pi_{W|B}$. The partition on $\mathcal{V}$ induced by $\equiv_{W|B}$ reduces the complexity of the problem associated with vector space. Instead of studying each vector individually, we can study one vector from each block. This single vector will serve as a representative vector for entire block and the characteristics of this individual vector are sufficient to provide full information about all the vectors in the block. The number of distinct indiscernibility blocks of a partition indicates the number of vectors sufficient to study characteristics of all the vectors in $\mathcal{V}$. If $\Pi_{W|B}$ consists of m blocks, then we write $|\Pi_{W|B}| = m$. We say $\Pi_{\mathcal{V}|B}$ is the canonical partition, note that each block of this partition consists of vectors with the same skeleton. For each $\mathcal{B} \subseteq B$, we associate a set of vectors as $\mathcal{C}_{\mathcal{B}} = \{v \in \mathcal{V} : S_{v|B} = \mathcal{B}\}$. Note that the sets $\mathcal{C}_{\mathcal{B}}$ form a partition on $\mathcal{V}$ which coincides with the canonical partition. Moreover, we say all the vectors in $\mathcal{V}$ are indiscernible if $W = \emptyset$.

In the sequel, if the basis is understood from the context, we denote the information system by I , the information map by $\mathcal{F}$, the skeleton v by S_v , the representation of u with respect to W by $\Gamma(u, W)$, and partition on $\mathcal{V}$ with respect to $W \subseteq \mathcal{V}$ by Π_W .

In the next result, we prove that the number of blocks in Π_W depends on the cardinality of W .

Theorem 3.1. For a finite subset $W \subseteq \mathcal{V}$, $1 \leq |\Pi_W| \leq 2^{|W|}$.

Proof. We prove the assertion by the following two cases.

Case 1: If $W = \emptyset$ then all the vectors in $\mathcal{V}$ are indiscernible, therefore $|\Pi_W| = 1$.

Case 2: Let $W = (w_1, w_2, \dots, w_s)$ be a non-empty ordered subset of $\mathcal{V}$. According to equation (3), each $v \in \mathcal{V}$ will have a representation $\Gamma_B(v, W)$ with $|W|$ entries. Note that each entry in the representation is either 0 or 1, so there are at most $2^{|W|}$ different representations. Therefore, $|\Pi_W| \leq 2^{|W|}$. $\square$

Now we prove that the upper bound in Theorem 3.1 is sharp.

Proposition 3.2. For a finite subset $W \subseteq \mathcal{V}$ with $|W| = s$, $|\Pi_W| = 2^s$ if the following conditions hold.

1. For $n = 1$, if $s = 1$ and $S_w = B$ for $w \in W$.
2. For $n \geq 2$ and $s \geq 2$, if $\cap_{i=1}^s S_{w_i} = \emptyset$ and $\cup_{i=1}^s S_{w_i} \subseteq B$ for all $w_i \in W$.

Proof. (1) Since the dimension of $\mathcal{V}$ is 1, so all the non-zero vectors v will have same skeleton and $S_v = B$. Suppose contrary that $s \geq 2$ then we have:

Case 1: If $\theta \in W$, then $\mathcal{F}(\theta, v) = 0$ for all $v \in \mathcal{V}$. For any non-zero vector $w \in W$, $\mathcal{F}(w, u) = 1$ for all $u \in \mathcal{V} \setminus \{\theta\}$. This implies $|\Pi_W| = 2 \neq 2^s$, so our supposition that $s \geq 2$ is wrong.

Case 2: If all the vectors in W are non-zero vectors, then $\mathcal{F}(\theta, w) = 0$ and $\mathcal{F}(v, w) = 1$ for all non-zero vectors v , which implies $|\Pi_W| = 2 \neq 2^s$, so our supposition that $s \geq 2$ is wrong.

Hence, $s = 1$. Now it is left to prove that $S_w = B$, for $w \in W$. Suppose contrary that $S_w \neq B$. Since $n = 1$, so $W = \{\theta\}$ which implies $|\Pi_W| = 1 \neq 2^s$. Therefore, our supposition is wrong, hence $S_w = B$ for $w \in W$.

(2) Suppose $W \subseteq \mathcal{V}$. Let $\cap_{i=1}^s S_{w_i} = \emptyset$ and $\cup_{i=1}^s S_{w_i} \subseteq B$ for all $w_i \in W$, then $\mathcal{F}(u, w) \neq \mathcal{F}(v, w)$ for all $u, v \in \mathcal{V}$ such that $S_u \neq S_v$. Since $|W| = s$, which implies there exist 2^s distinct representations. Hence $|\Pi_W| = 2^s$. $\square$

In the following result, we provide the condition for which the upper bound in Theorem 3.1 is not attained.

Proposition 3.3. For a finite subset $W \subseteq \mathcal{V}$, $|\Pi_W| < 2^{|W|}$ if the following conditions hold.

1. If $n = 1$ and $|W| \geq 2$.
2. If $n \geq 2$ and $S_{w_i} \subseteq S_{w_j}$ for some $w_i, w_j \in W$.

Proof. (1) Suppose $n = 1$ and $|W| \geq 2$, then we have:

Case 1: If $\theta \in W$ then $\mathcal{F}(\theta, v) = 0$, for all $v \in \mathcal{V}$ and for any non-zero vector $w \in W$, $\mathcal{F}(w, u) = 1$ for all $u \in \mathcal{V} \setminus \{\theta\}$. This implies $|\Pi_W| = 2 < 2^s$.

Case 2: If all the vectors in W are non-zero vectors, then $\mathcal{F}(\theta, w) = 0$ and $\mathcal{F}(v, w) = 1$ for all non-zero vectors v , which implies $|\Pi_W| = 2 < 2^s$.

(2) Let $W = (w_1, w_2, \dots, w_s)$ be an ordered subset of $\mathcal{V}$. Suppose $S_{w_i} = S_{w_j}$, where $i \neq j$ and $i, j \in \{1, 2, \dots, s\}$, then $\mathcal{F}(v, w_i) = \mathcal{F}(v, w_j)$ for all $v \in \mathcal{V}$, which implies the i_{th} and j_{th} entry in the representation $\Gamma_B(u, W)$, for any $u \in \mathcal{V}$, are the same. Since each entry in the representation is either 0 or 1, so in $\Gamma_B(u, W)$ the i_{th} and j_{th} entries are both 1 or 0. This implies $|\Pi_W| < 2^{|W|}$. Now suppose $S_{w_i} \subset S_{w_j}$, then for any $v \in \mathcal{V}$ if $\mathcal{F}(v, w_i) = 1$ implies $\mathcal{F}(v, w_j) = 1$. This implies there does not exist any vector $u \in \mathcal{V}$, which has 1 at i_{th} place in its representation and 0 at j_{th} place in its representation $\Gamma_B(u, W)$. Hence, $|\Pi_W| < 2^{|W|}$. $\square$

In the next result, we prove that the number of partition blocks in a canonical partition depends on the dimension n of $\mathcal{V}$ and show that partition induced by any superset of a basis B is same as the canonical partition.

Proposition 3.4. For a basis B of $\mathcal{V}$, $|\Pi_{\mathcal{V}}| = 2^n$. Moreover, if $B \subseteq W$ then $\Pi_W = \Pi_{\mathcal{V}}$.

Proof. Since each block of $\Pi_{\mathcal{V}}$ consists of vectors having the same skeleton so depending upon n , there exist $\binom{n}{k}$ blocks each consisting of vectors having k basis vectors in their skeleton, where $0 \leq k \leq n$. Therefore, $|\Pi_{\mathcal{V}}| = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n$.

Suppose $B \subseteq W$, to prove that $\Pi_W = \Pi_{\mathcal{V}}$ we have:

Case 1: Let $W = B$ then from Proposition 3.2, $|\Pi_B| = 2^n$. Since $|\Pi_{\mathcal{V}}| = 2^n$ and each block of $\Pi_{\mathcal{V}}$ consists of vectors having the same skeleton, which implies there exist only one such partition of this cardinality. Hence, $\Pi_W = \Pi_{\mathcal{V}}$.

Case 2: Let $W \subseteq \mathcal{V}$ such that $B \subset W$. From Case 1, $\Pi_B = \Pi_{\mathcal{V}}$, this implies for any two vectors with distinct skeletons have distinct entries in at least one of the places in the first n entries of their representation. This implies all the vectors which are discernible with respect to B are also discernible with respect to W . Moreover suppose $S_u = S_v$, then from Case 1 $u \equiv_B v$ and so $\mathcal{F}(u, w) = \mathcal{F}(v, w)$ for any $w \in W \setminus B$, this implies all the vectors which are indiscernible with respect to B are also indiscernible with respect to W . Therefore, $\Pi_B = \Pi_W$ which implies from Case 1 that $\Pi_W = \Pi_{\mathcal{V}}$. $\square$

In the case of $\mathcal{V}$ over F with $|F| = 2$, the canonical partition of $\mathcal{V}$ has a unique partition structure, because all the vectors have unique skeleton as we demonstrate in the following example.

Example 3.4. For $\mathcal{V}$ over $\mathbb{Z}_2$ with $n = 2$ and $B = \{(1, 0), (0, 1)\}$ then $S_{(1,0)} = \{(1, 0)\}$, $S_{(0,1)} = \{(0, 1)\}$ and $S_{(1,1)} = \{(1, 0), (0, 1)\}$.

The following result is a special case of Proposition 3.4.

Corollary 3.1. For $\mathcal{V}$ over a field F with $|F| = 2$, if $W \subseteq \mathcal{V}$ and $B \subseteq W$ then $\Pi_{W|B} = v_1 | v_2 | \cdots | v_s$, where $|\mathcal{V}| = s$.

Note that for a vector space $\mathcal{V}$ over an infinite field F , the cardinality of each indiscernibility block of $\Pi_{\mathcal{V}}$ is infinite except for the indiscernibility block containing the null vector. In the case of finite fields, the cardinality of each indiscernibility block of $\Pi_{\mathcal{V}}$ depends on the cardinality of the skeleton of vectors in it and the cardinality of the field F . If $[v]$ is an indiscernibility block of $\Pi_{\mathcal{V}}$ consisting of vectors having skeleton of cardinality l , where $0 \leq l \leq n$, then $|[v]| = (|F| - 1)^l$.

Let we define the set consisting of vectors having same number of basis vectors in their skeleton as $T_i = \{v \in \mathcal{V} : |S_v| = i\}$, where $i \in \{1, 2, \dots, n\}$. In Proposition 3.4, we have discussed the partition structure induced on $\mathcal{V}$ with respect to a subset W of $\mathcal{V}$ which contains basis B in it. Now, we provide the partition structure induced by $\mathcal{V} \setminus B$ on $\mathcal{V}$.

Theorem 3.2. For I_B associated with $\mathcal{V}$ of dimension n , if $W = \mathcal{V} \setminus B$ then:

1. If $n = 1$ and $|F| = 2$, then all the vectors in $\mathcal{V}$ are indiscernible.
2. If $n \geq 2$ and $|F| = 2$, then $\Pi_W = \theta | v_1 | v_2 | \cdots | v_s | T_{n-1} \cup T_n$, where $\cup_{i=1}^{n-2} T_i = \{v_1, v_2, \dots, v_s\}$.
3. If $|F| \geq 3$, then $\Pi_W = \Pi_{\mathcal{V}}$.

Proof. Let $W = \mathcal{V} \setminus B$, then:

- (1) For $|F| = 2$ and $n = 1$, $|\mathcal{V}| = 2$ and there exist only one basis $B = \{b\}$. Let $W = \mathcal{V} \setminus B$, then $W = \{\theta\}$, which implies $\forall v \in \mathcal{V}$ are indiscernible with respect to $W = \mathcal{V} \setminus B$.
- (2) Let $|F| = 2$ and $n \geq 2$, then we know that all the vectors have a unique skeleton. Suppose $W = \mathcal{V} \setminus B$, then $|S_w| \geq 2$ for every non-zero vector in W . Let $w \in W$ with $|S_w| \leq n - 2$, then there exists at least one $v \in \mathcal{V}$ with $|S_v| \leq n - 2$ such that $S_v \cap S_w = \emptyset$ which implies $\mathcal{F}(v, w) = 0$ and for all $\hat{w} \in W$ $\mathcal{F}(\hat{w}, v) = 1$. So, each vector $v \in \mathcal{V}$ with $|S_v| \leq n - 2$, has a unique representation with respect to W . Therefore, $[v_j]_W = \{v_j\}$ for each $v_j \in \cup_{i=1}^{n-2} T_i$. Moreover, the zero vector also has a unique representation because it contains all zeros in its representation. Now suppose $u \in T_i$ for some $2 \leq i \leq n$ and $v \in T_{n-1} \cup T_n$, then $|S_w| = i$ and $|S_v| \geq n - 1$, which implies $|S_w \cap S_v| \geq 1$, so $\mathcal{F}(v, w) = 1$ and $u \equiv_W v$, $\forall u, v \in T_{n-1} \cup T_n$. Therefore, $\Pi_W = \theta | v_1 | v_2 | \cdots | v_s | T_{n-1} \cup T_n$.
- (3) Let $|F| \geq 3$, then for each basis vector $b_i \in B$ there exist at least one vector $kb_i \in W$, where $k \in F \setminus \{0\}$, such that $S_{b_i} = S_{kb_i}$, which implies $\mathcal{F}(v, b_i) = \mathcal{F}(v, kb_i)$, $\forall v \in \mathcal{V}$. Hence from Proposition 3.4, $\Pi_W = \Pi_{\mathcal{V}}$. $\square$

The indiscernibility of two vectors depends on the choice of $W \subseteq \mathcal{V}$. In the following result, we discuss which vectors are indiscernible with respect to collection of all those vectors having skeleton of same cardinality.

Proposition 3.5. If $W = T_i$, then $u \equiv_W v$ for all $u, v \in \cup_{m=n-i+1}^n T_m$, where $1 \leq i \leq n$.

Proof. Let $W = T_i$ for some $i \in \{1, 2, \dots, n\}$, $v \in T_m$ for some $n - i + 1 \leq m \leq n$, then $|S_w| = i$ and $|S_v| \geq n - i + 1$. Since dimension of $\mathcal{V}$ is n , so $|S_w \cap S_v| \geq 1$, which implies $\mathcal{F}(v, w) = 1$. Therefore, $u \equiv_W v$ for all $u, v \in \cup_{m=n-i+1}^n T_m$. Moreover, for any vector $\hat{u} \in \mathcal{V}$ with $|S_{\hat{u}}| < n - i + 1$ there exist at least one $w \in W$ such that $S_{\hat{u}} \cap S_w = \emptyset$, which implies $\mathcal{F}(\hat{u}, w) = 0$. Therefore, $u \equiv_W v$, $\forall u, v \in \cup_{m=n-i+1}^n T_m$. $\square$

The next result follows from Proposition 3.5 and shows the partition structure induced on $\mathcal{V}$ by a set of vectors having skeleton equal to basis.

Corollary 3.2. For $W \subseteq T_n$, $|\Pi_W| = 2$ and $\Pi_W = \theta | \mathcal{V} \setminus \theta$.

In this section thus far, we have discussed the indiscernibility of vectors. Now, we discuss the indiscernibility of two subsets of a vector space. In [6], it is defined that the two feature subsets of an information system are indiscernible if they induce same partition. Analogously, we define indiscernibility of two subsets $W_1, W_2 \subseteq \mathcal{V}$ as:

$$W_1 \approx W_2 \Leftrightarrow \Pi_{W_1} = \Pi_{W_2}.$$

Since each subset $W \subseteq \mathcal{V}$ induces a partition on $\mathcal{V}$, identifying indiscernible subsets of $\mathcal{V}$ and selecting one subset from each such collection is sufficient to study. Moreover, a subset $W \subseteq \mathcal{V}$ may contain some vectors which do not contribute to the partition induced by W as removal of such vectors from W does not change the partition induced by W , we refer these vectors as dispensable vectors of W .

Definition 3.4. Let $W \subseteq \mathcal{V}$, we say a subset $X \subseteq W$ is a dispensable set of W , if $\Pi_W = \Pi_{W \setminus X}$ and $\Pi_W \neq \Pi_{W \setminus Y}$ for all $X \subset Y$.

Example 3.5. For $\mathcal{V}$ over $\mathbb{Z}_3$ with $n = 2$ and $B = \{(1, 0), (0, 2)\}$, if $W = \{(1, 0), (0, 1), (1, 2)\}$ then $\text{Disp}(W) = \{(1, 2)\}$.

The following results illustrate the conditions under which two subsets of $\mathcal{V}$ induce the same partition on $\mathcal{V}$.

Proposition 3.6. For $W_i, W_j \subseteq \mathcal{V}$, $W_i \approx W_j$, if for each $w_i \in W_i \setminus \text{Disp}(W_i)$ there exists $w_j \in W_j \setminus \text{Disp}(W_j)$ such that $S_{w_i} = S_{w_j}$.

Proof. Suppose $W_i, W_j \subseteq \mathcal{V}$ and for each $w_i \in W_i \setminus \text{Disp}(W_i)$ there exists $w_j \in W_j \setminus \text{Disp}(W_j)$ such that $S_{w_i} = S_{w_j}$, then $\mathcal{F}(u, w_i) = \mathcal{F}(u, w_j)$ for each $u \in \mathcal{V}$ which implies $\Pi_{W_i} = \Pi_{W_j}$. Hence, $W_i \approx W_j$. $\square$

The next result follows from Proposition 3.6.

Corollary 3.3. If $W_1, W_2 \subseteq \mathcal{V}$ contains vectors from each block of $\Pi_{\mathcal{V}}$ then $\Pi_{W_1} = \Pi_{W_2}$.

In machine learning and data science, removing dispensable features from data and keeping the indispensable set of features is referred as feature selection. These indispensable set of features are referred as reducts. Note that there may exist more than one reducts of an information system and the set of features which belongs to every reduct is called the core. Now, we investigate the reducts and the core of I_B . A subset R_{I_B} of $\mathcal{V}$ is reduct of I_B , if $\Pi_{R_{I_B}} = \Pi_{\mathcal{V}}$ and $\Pi_X \neq \Pi_{\mathcal{V}}$, $\forall X \subset R_{I_B}$, and $C_{I_B} \subseteq \mathcal{V}$ is the core of I_B , if $\Pi_{\mathcal{V}} \neq \Pi_{\mathcal{V} \setminus \{c\}}$ for all $c \in C_{I_B}$. If the basis is understood from the context, we denote reduct and core of I by R and C , respectively.

In the following result, we provide the relation between basis vectors $b \in B$ and a reduct of I_B .

Theorem 3.3. For I_B , $R_{I_B} \subseteq \mathcal{V}$ is reduct if $|R_{I_B} \cap [b]_{\mathcal{V}}| = 1$ for all $b \in B$, and $R_{I_B} \cap (\cup_{i=2}^n T_i \cup \{\theta\}) = \emptyset$.

Proof. Let $R_{I_B} \subseteq \mathcal{V}$ such that $|R_{I_B} \cap [b]_{\mathcal{V}}| = 1$ for all $b \in B$, and $R_{I_B} \cap (\cup_{i=2}^n T_i \cup \{\theta\}) = \emptyset$. To prove that R_{I_B} is a reduct, we have to show that $\Pi_{R_{I_B}} = \Pi_{\mathcal{V}}$ and $\Pi_X \neq \Pi_{\mathcal{V}}$ for every $X \subset R_{I_B}$. From Proposition 3.4, $\Pi_{\mathcal{V}} = \Pi_B$ and for each $X \subset B$ $\Pi_X \neq \Pi_{\mathcal{V}}$. Since $B \cap [b]_{\mathcal{V}} = 1$ and $|S_v| = 1$ for all $v \in B$, which implies $B \cap (\cup_{i=2}^n T_i \cup \{\theta\}) = \emptyset$ and B is a reduct. Hence, R_{I_B} is reduct of I_B . $\square$

The above result shows that the reducts of an information system I_B depends on the underlying basis B . Moreover, it is evident that each reduct is a basis, whereas every basis is not a reduct.

Example 3.6. For I_B associated with $\mathcal{V}$ over $\mathbb{Z}_3$ with $n = 2$ and $B = \{(1, 0), (0, 2)\}$. The set $W = \{(1, 0), (1, 1)\}$ is also a basis of $\mathcal{V}$ but $\Pi_{\mathcal{V}} \neq \Pi_W$, so W is not a reduct of I_B .

Let $R_{\mathcal{V}|I_B}$ denote the collection of all reducts of information system I_B of $\mathcal{V}$. Now we prove that the number of reducts depends on the dimension and cardinality of the field of vector space.

Proposition 3.7. For $\mathcal{V}$ over field F , $|R_{\mathcal{V}|I_B}| = (|F| - 1)^n$.

Proof. From Theorem 3.3, we have $|R_{I_B} \cap [b]_{\mathcal{V}}| = 1$ for all $b \in B$ and $|R_{I_B}| = n$. Since $|[b]_{\mathcal{V}}| = |F| - 1$, which implies $|R_{\mathcal{V}|I_B}| = (|F| - 1)^n$. $\square$

The next result follows from Proposition 3.7.

Corollary 3.4. For $\mathcal{V}$ over an infinite field F , there exist an infinite number of reducts of I .

Algorithm 1 Reducts of an information system I .

Input: $R = \{X_j \subseteq \mathcal{V} : 1 \leq j \leq s = 2^{|\mathcal{V}|}\}$ ▷ Collection of all subsets of $\mathcal{V}$.
Output: R ▷ Collection of all reducts of I associated with $\mathcal{V}$.
Initialize: $j = 1$
for $j \leq s$ **do**
 if $\Pi_{X_j} \neq \Pi_{\mathcal{V}}$ **then**
 $R = R \setminus X_j$
 $j = j + 1$
 else
 $j = j + 1$
 end if
end for
 $R = \{A_l : 1 \leq l \leq t\}$ ▷ R updated from the above for loop
 $l = 1$
for $l \leq t$ **do**
 $H_l = \{Y_i \subset A_l : 1 \leq i \leq r\}$ ▷ Collection of all proper subsets of $A_l \in R$
 if $\Pi_{Y_i} = \Pi_{\mathcal{V}}$, for at least one $i \in \{1, 2, \dots, r\}$ **then**
 $R = R \setminus A_l$
 $l = l + 1$
 else
 $l = l + 1$
 end if
end for
Return R

The Algorithm 1 computes all reducts of I associated with $\mathcal{V}$.

Now, we determine the core of I associated with $\mathcal{V}$. In the following result, we show the relation between the core and reduct of an information system associated with vector spaces over field F with $|F| = 2$.

Theorem 3.4. For an information system I associated with $\mathcal{V}$ over F with $|F| = 2$, reduct coincides with the core of I .

Proof. Let $B = \{b_1, b_2, \dots, b_n\}$ and $|F| = 2$, then from Theorem 3.3 and Proposition 3.7 B is the only reduct of I . Now from Proposition 3.4, $\Pi_{\mathcal{V}} \neq \Pi_X$ for any $X \subset B$ which implies $\Pi_{\mathcal{V}} \neq \Pi_{\mathcal{V} \setminus \{b_i\}}$. Therefore, all the basis vectors belong to the core of I . Hence $R = C$. $\square$

The reducts and core of an information system are sufficient to preserve the partition structure of data, although there does not necessarily exist the core for every information system.

Remark 3.1. The core of the information systems associated with $\mathcal{V}$ over field F with $|F| \geq 3$ is an empty set because $|[b]_{\mathcal{V}}| \geq 2$ for all $b \in B$, which implies $\Pi_{\mathcal{V}} = \Pi_{\mathcal{V} \setminus \{v\}}$ for any $v \in \cup_{b \in B} [b]_{\mathcal{V}}$, and from Proposition 3.4, $\Pi_{\mathcal{V}} = \Pi_{\mathcal{V} \setminus \{u\}}$ for any $u \in \cup_{i=2}^n T_i$. Therefore, there exists no single vector $w \in \mathcal{V}$ such that $\Pi_{\mathcal{V}} \neq \Pi_{\mathcal{V} \setminus \{w\}}$.

Example 3.7. For $\mathcal{V}$ over $\mathbb{Z}_3$ with $n = 2$ and $B = \{(1, 0), (0, 2)\}$, $R_{\mathcal{V}|B} = \{\{(1, 0), (0, 1)\}, \{(2, 0), (0, 2)\}, \{(1, 0), (0, 2)\}, \{(2, 0), (0, 1)\}\}$ and $C_{I_B} = \emptyset$.

Now, we discuss the condition under which two subsets of $\mathcal{V}$ induce distinct partitions on $\mathcal{V}$ with respect to a basis B of $\mathcal{V}$.

Proposition 3.8. Let $W_1, W_2 \subseteq \mathcal{V}$ then $\Pi_{W_1} \neq \Pi_{W_2}$, if there exist at least one $w \in W_1$ such that $S_w \neq S_{\hat{w}}$ for all $\hat{w} \in W_2 \setminus \text{Disp}(W_2)$.

Proof. Suppose $W_1, W_2 \subseteq \mathcal{V}$ and let $w \in W_1$ such that $S_w \neq S_{\hat{w}}$, for all $\hat{w} \in W_2 \setminus \text{Disp}(W_2)$. Now let there exist $u, v \in \mathcal{V}$ such that $u \equiv_{W_1} v$, which implies $\mathcal{F}(u, w) = \mathcal{F}(v, w)$ for all $w \in W_1$. Since $S_w \neq S_{\hat{w}}$ so $\mathcal{F}(u, \hat{w}) \neq \mathcal{F}(v, \hat{w})$, which implies $u \not\equiv_{W_2} v$. Hence, $\Pi_{W_1} \neq \Pi_{W_2}$. $\square$

Note that for any two basis B_1 and B_2 of $\mathcal{V}$, there may exist a subset W of $\mathcal{V}$ which induce distinct partitions on $\mathcal{V}$ with respect to distinct basis of $\mathcal{V}$. In the following result, we provide the condition under which a subset W of $\mathcal{V}$ induces distinct partitions on $\mathcal{V}$ with respect to distinct basis.

Proposition 3.9. For two distinct basis B and $\hat{B}$ of $\mathcal{V}$, let W be a non-empty subset of $\mathcal{V}$ and $W \neq \{\emptyset\}$, then $\Pi_{W|B} \neq \Pi_{W|\hat{B}}$, if for some $b \in B$ there does not exist $\hat{b} \in \hat{B}$ such that $b = k\hat{b}$, where $k \in F \setminus \{0\}$.

Proof. Let for a basis vector $b \in B$ there does not exist any $\hat{b} \in \hat{B}$ such that $b = k\hat{b}$, which implies there exist at least one vector $v \in \mathcal{V}$ such that $S_{v|B} \neq S_{v|\hat{B}}$. Since $S_{v|B} \neq S_{v|\hat{B}}$, which implies there exist at least two distinct vectors u and $\hat{u}$ such that $u \equiv_{v|B} \hat{u}$ and $u \not\equiv_{v|\hat{B}} \hat{u}$. Hence, $\Pi_{W|B} \neq \Pi_{W|\hat{B}}$. $\square$

From the results established thus far, we observe that the partition induced by $W \subseteq \mathcal{V}$ on $\mathcal{V}$ depends on the skeleton of vectors in W and from Theorem 3.1, it is clear that there may exist two distinct partitions of same cardinality. Two partitions having the same cardinality may or may not have corresponding partition blocks of same cardinalities. The former case corresponds to size-isomorphism of partitions [34]. We now define size-isomorphism between partitions induced on $\mathcal{V}$.

Definition 3.5. Two partitions Π_{W_1}, Π_{W_2} are size-isomorphic, if $|\Pi_{W_1}| = |\Pi_{W_2}|$ and there exists a bijective map σ between the partitions such that $|\pi_{W_1}| = |\sigma(\pi_{W_1})|$ for all blocks $\pi_{W_1} \in \Pi_{W_1}$.

Note that there exist various subsets of $\mathcal{V}$ for which their corresponding partitions are size-isomorphic, as we illustrate in the following example.

Example 3.8. For $\mathcal{V}$ over $\mathbb{Z}_3$ with $n = 2$ and $B = \{(1, 0), (0, 1)\}$, the partitions $\Pi_{\{(1, 0)\}}$ and $\Pi_{\{(0, 1)\}}$ are size-isomorphic. Similarly, if $W_1 = \{(1, 0), (1, 1)\}$ and $W_2 = \{(0, 1)\}$ then the corresponding partitions Π_{W_1} and Π_{W_2} are size-isomorphic.

In the next result, we prove that any two canonical partitions of a vector space $\mathcal{V}$ are size-isomorphic partitions.

Theorem 3.5. For two distinct basis B and $\hat{B}$ of $\mathcal{V}$, the canonical partitions $\Pi_{\mathcal{V}|B}$ and $\Pi_{\mathcal{V}|\hat{B}}$ are size-isomorphic.

Proof. Suppose $B, \hat{B}$ are two distinct basis of $\mathcal{V}$, then from Proposition 3.4, $|\Pi_{\mathcal{V}|B}| = |\Pi_{\mathcal{V}|\hat{B}}| = 2^n$. Let $\sigma : \Pi_{\mathcal{V}|B} \rightarrow \Pi_{\mathcal{V}|\hat{B}}$ be defined by $\sigma([v]_{\mathcal{V}|B}) = [\hat{v}]_{\mathcal{V}|\hat{B}}$, where $|S_v| = |S_{\hat{v}}|$. Clearly σ is a bijective map because for any two distinct basis of $\mathcal{V}$, the number of vectors having skeleton of cardinality l is same where $l \in \{0, 1, 2, \dots, n\}$. Now, we prove that $|[v]_{\mathcal{V}|B}| = |[\hat{v}]_{\mathcal{V}|\hat{B}}|$. Since each block in a canonical partition consists of vectors with same skeleton, which implies that $|[v]_{\mathcal{V}|B}| = |[\hat{v}]_{\mathcal{V}|\hat{B}}|$ and concludes the proof. $\square$

The defined indiscernibility relation provides a framework to identify similarities between feature vectors and induces partition on feature space. The partitioning of feature space into groups reduces complexity of the data under study.

4. Dependency measure, maximum partitioners and lattice structures

In this section, we discuss the measure of dependency of two subsets of a vector space and define different lattice structures associated with $\mathcal{V}$.

In [6], it is shown that the inclusion relation between two feature subsets of an information system corresponds to partial order relation between their corresponding partitions. For two partitions Π_1, Π_2 induced on a set, the partition Π_1 is called coarser partition than Π_2 or Π_2 is called finer than Π_1 , if $|\Pi_1| \leq |\Pi_2|$ and each block of Π_2 is contained in a block of Π_1 . Analogously, we define inclusion relation between two subsets W_1, W_2 of $\mathcal{V}$ and a partial order between the partitions induced by them as

$$W_1 \subseteq W_2 : \Leftrightarrow \Pi_{W_2} \leq \Pi_{W_1}, \quad (5)$$

where Π_{W_2} and Π_{W_1} are finer and coarser partitions, respectively.

The next result provides the conditions under which finest and coarsest partitions of $\mathcal{V}$ are obtained.

Proposition 4.1. For I_B , if $B \subseteq W$ then Π_W is the finest partition of $\mathcal{V}$. Moreover, $\Pi_{\{\emptyset\}}$ and $\Pi_{\emptyset}$ are the coarsest partitions of $\mathcal{V}$.

In [6], an indiscernibility partition lattice is defined for an information system. Analogously, we define an indiscernibility partition poset $\mathfrak{L}_\Pi = (\Lambda_W, \leq)$, where $\Lambda_W = \{\Pi_W \mid W \subseteq \mathcal{V}\}$. The indiscernibility partition lattice demonstrates the hierarchical relationship between all the possible partitions induced by subsets of $W \subseteq \mathcal{V}$ in context of the level of granularity of partitions.

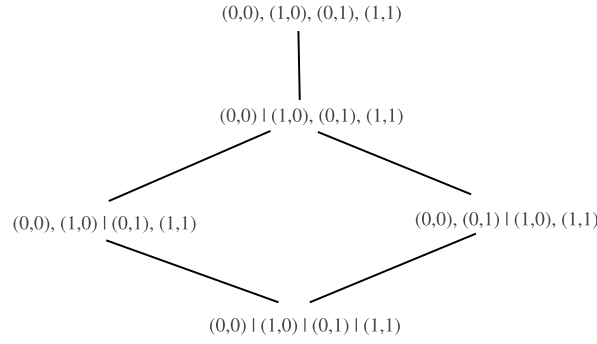
Example 4.1. For $\mathcal{V}$ over $\mathbb{Z}_2$ with $n = 2$ and $B = \{(1, 0), (0, 1)\}$. Since each $W \subseteq \mathcal{V}$ corresponds to a partition, so we have:

$$\begin{aligned} \Pi_{\emptyset} &= \Pi_{\{(0,0)\}} = (0, 0), (1, 0), (0, 1), (1, 1), \\ \Pi_{\{(1,0)\}} &= \Pi_{\{(0,0),(1,0)\}} = \Pi_{\{(1,0),(1,1)\}} = \Pi_{\{(0,0),(1,0),(1,1)\}} = (0, 0), (0, 1) \mid (1, 0), (1, 1), \\ \Pi_{\{(0,1)\}} &= \Pi_{\{(0,0),(0,1)\}} = \Pi_{\{(0,1),(1,1)\}} = \Pi_{\{(0,0),(1,0),(1,1)\}} = (0, 0), (1, 0) \mid (0, 1), (1, 1), \\ \Pi_{\{(1,1)\}} &= \Pi_{\{(0,0),(1,1)\}} = (0, 0) \mid (1, 0), (0, 1), (1, 1), \\ \Pi_{\{(1,0),(1,0)\}} &= \Pi_{\{(0,0),(1,0),(0,1)\}} = \Pi_{\{(1,0),(0,1),(1,1)\}} = \Pi_{\{(0,0),(1,0),(0,1),(1,1)\}} = (0, 0), (0, 1) \mid (1, 0), (1, 1). \end{aligned}$$

So, there are a total of 5 distinct partition of $\mathcal{V}$. The associated partition lattice is shown in Fig. 1.

For $W \subseteq \mathcal{V}$, we define an indiscernibility partition poset with respect to W as $\mathfrak{L}_W = (\Lambda_{\hat{W}}, \leq)$, where $\Lambda_{\hat{W}} = \{\Pi_{\hat{W}} \mid \hat{W} \subseteq W\}$ is the collection of all distinct partitions of $\mathcal{V}$ induced by subsets of W . Note that for any two subsets W_1, W_2 of $\mathcal{V}$, if Π_{W_1} and Π_{W_2} are size-isomorphic partitions then their corresponding posets $\mathfrak{L}_{W_1}$ and $\mathfrak{L}_{W_2}$ are isomorphic.

The relation between two indiscernibility partitions is related to some notions of rough set, such as positive region and dependency measure [6]. The positive region between two indiscernibility partitions corresponds to set of those objects which are classified

Fig. 1. $\mathfrak{L}_\Pi$.

indiscernible with respect to both indiscernibility partitions. We define the positive region and dependency measure of W_2 with respect to W_1 respectively as

$$POS_{W_1}(W_2) = \{v \in \mathcal{V} : [v]_{W_1} \subseteq [v]_{W_2}\}, \quad (6)$$

$$\Upsilon_{W_1}(W_2) = \frac{|POS_{W_1}(W_2)|}{|\mathcal{V}|}. \quad (7)$$

Note that if $\Pi_{W_1} \leq \Pi_{W_2}$, then $POS_{W_1}(W_2) = \mathcal{V}$ and $\Upsilon_{W_1}(W_2) = 1$.

Now, in the following results, we investigate positive regions between subsets of $\mathcal{V}$. In the next result, we provide description of a subset of $\mathcal{V}$ with respect to which the positive region of a proper subset of a basis B is a non-empty proper subset of $\mathcal{V}$.

Proposition 4.2. For non-empty subsets W_1, W_2 of $\mathcal{V}$, if $W_1 = \{b_1, b_2, \dots, b_s\} \subset B$ with $|W_1| \geq 2$ and $W_2 = \{v : v = \sum_{i=1}^s k_i b_i, k_i \in F \setminus \{0\}\}$, then $POS_{W_2}(W_1) = \{v \in \mathcal{V} : S_v \subseteq B \setminus W_1\}$.

Proof. Let $W_1 = \{b_1, b_2, \dots, b_s\} \subset B$ with $|W_1| \geq 2$ then from Theorem 3.1, $|\Pi_{W_1}| = 2^s$ and for all $u \in \mathcal{V}$ such that $S_u \cap S_{b_i} = \emptyset$, for all $i \in \{1, 2, \dots, s\}$, are indiscernible with respect to W_1 which implies $S_u \subseteq B \setminus W_1$ and $u \in [\theta]_{W_1}$. Now, let $W_2 = \{v : v = \sum_{i=1}^s k_i b_i, k_i \in F \setminus \{0\}\}$ then from Theorem 3.1, $|\Pi_{W_2}| = 2$. Since $S_v = \{b_1, b_2, \dots, b_s\}$ for all $v \in W_2$, therefore for all $\hat{v} \in \mathcal{V}$ with $S_{\hat{v}} \subseteq B \setminus S_v$ are indiscernible with respect to W_2 , because $\mathcal{F}(v, \hat{v}) = 0$, which implies $\hat{v} \in [\theta]_{W_2}$. Since $S_v \cap S_{b_i} \neq \emptyset$ for all $i \in \{1, 2, \dots, s\}$, so $[\theta]_{W_1} = [\theta]_{W_2}$. Therefore, $POS_{W_2}(W_1) = \{v \in \mathcal{V} : S_v \subseteq B \setminus W_1\}$.

In the following result, we discuss the positive regions of two distinct subsets of a basis B of $\mathcal{V}$.

Proposition 4.3. For non-empty subsets W_1, W_2 of B , if $W_1 \neq W_2$ then $POS_{W_1}(W_2) = \emptyset$ and $POS_{W_2}(W_1) = \emptyset$.

Proof. Let W_1, W_2 be non-empty subsets of B and $W_1 \neq W_2$, then for each $b_i \in W_1$ and $b_j \in W_2$, $S_{b_i} \cap S_{b_j} = \emptyset$, which implies that for each pair of vectors $u, \hat{u} \in \mathcal{V}$, if $u \equiv_{W_1} \hat{u}$ then $u \not\equiv_{W_2} \hat{u}$. Similarly, for $v, \hat{v} \in \mathcal{V}$, if $v \equiv_{W_2} \hat{v}$ then $v \not\equiv_{W_1} \hat{v}$. Therefore, $POS_{W_1}(W_2) = \emptyset$ and $POS_{W_2}(W_1) = \emptyset$. $\square$

Let $B_i = \{kb_i : k \in F \setminus \{0\}\}$ be a set of vectors associated with each basis vector b_i . In the following result, we discuss the positive region of different subsets of $\mathcal{V}$ with respect to B_i .

Proposition 4.4. If $W_1 \subseteq \mathcal{V}$ and $W_2 = B_l$ where $l \in \{1, 2, \dots, n\}$, then

$$POS_{W_2}(W_1) = \begin{cases} \emptyset, & \text{if } W_1 = B \\ \emptyset, & \text{if } W_1 = T_k, k \in \{1, 2, \dots, n-1\}; \\ \emptyset, & \text{if } W_1 = B_k, l \neq k; \\ \{v \in \mathcal{V} : b_l \in S_v\}, & \text{if } W_1 \subseteq T_n. \end{cases}$$

Proof. Let $W_2 = B_l$ then Π_{B_l} consist of two indiscernibility blocks, one block consists of vectors v such that $b_l \in S_v$ and the second block consists of vectors v such that $b_l \notin S_v$.

(1) The proof follows from Corollary 4.1.

(2) Let $W_1 = T_k$, if $k = 1$ then from Theorem 3.4, $\Pi_{W_1} = \Pi_B$ which implies $\forall v \in \mathcal{V}$, $[v]_{W_2}$ is not a subset of any block of Π_{T_1} , hence $POS_{B_l}(T_1) = \emptyset$. Similarly for $k \in \{2, \dots, n-1\}$, $|\Pi_{T_k}| \geq 3$ which implies $\Pi_{T_k} < \Pi_{B_l}$, hence $POS_{B_l}(T_k) = \emptyset$.

(3) Let $W_2 = B_k$ then Π_{B_k} consist of two indiscernibility blocks π_1 and π_2 , such that $b_k \in S_u$ for all $u \in \pi_1$ whereas $b_k \notin S_u$ for all $u \in \pi_2$. Since $l \neq k$ which implies $[v]_{B_l} \not\subseteq [v]_{B_k}$, $\forall v \in \mathcal{V}$, hence $POS_{B_l}(B_k) = \emptyset$.

(4) Let $W_2 \subseteq T_n$, then from Theorem 3.1, $|\Pi_{T_n}| = 2$ and all the vectors with non-empty skeleton are indiscernible, which implies that the indiscernibility block of Π_{B_l} consisting of vectors v with $b_l \in S_v$ is contained in the block of Π_{T_n} which consists of vectors having a non-empty skeleton. Hence, $POS_{B_l}(T_n) = \{v \in \mathcal{V} : b_l \in S_v\}$. $\square$

Proposition 4.5. If $W_1, W_2 \subseteq \mathcal{V}$ then

$$POS_{W_1}(W_2) = \begin{cases} \mathcal{V}, & \text{if } W_1 = B, W_2 \subseteq \mathcal{V}; \\ \mathcal{V}, & \text{if } W_1 = T_1, W_2 \subseteq \mathcal{V}; \\ \emptyset, & \text{if } W_1 \subseteq T_n, W_2 \subseteq \mathcal{V} \setminus (\{\theta\} \cup T_n). \end{cases}$$

Proof. (1) From Corollary 4.1, Π_B is the finest partition, so $\Pi_B \leq \Pi_{W_2}$ for any $W_2 \subseteq \mathcal{V}$, hence $POS_{W_1}(W_2) = \mathcal{V}$.

(2) Since $\Pi_{T_1} = \Pi_B$ then from Proposition 4.5 (1), $\Pi_{T_1} \leq \Pi_{W_2}$, hence $POS_{W_1}(W_2) = \mathcal{V}$.

(3) From Corollary 4.1, $\Pi_{W_1} = \theta \mid \mathcal{V} \setminus \{\theta\}$, which implies $\Pi_{W_2} < \Pi_{W_1}$ for any $W_2 \subseteq \mathcal{V} \setminus (\{\theta\} \cup T_n)$. Therefore, $POS_{W_1}(W_2) = \emptyset$. $\square$

Note that for $W_1, W_2 \subseteq \mathcal{V}$, if partition induced by W_1 and W_2 is same, then $POS_{W_1}(W_2) = \mathcal{V}$ and $\gamma_{W_1}(W_2) = 1$. Moreover, for $\mathcal{V}$ over F with $|F| = 2$, if $W_1 \cap W_2 = \emptyset$ then $POS_{W_1}(W_2) = POS_{W_2}(W_1) = \emptyset$.

Let $[W]_{\approx}$ consist of all those subsets of $\mathcal{V}$ which are indiscernible to W . We say $Max(W) = \bigcup [W]_{\approx}$ is the maximum partitioner of W . Note that $Max(W)$ is the largest set in $[W]_{\approx}$ and it is unique [6]. Moreover, we say a set $X \in [W]_{\approx}$ is called a minimum partitioner of W , denoted by $Min(W)$, if $\Pi_X = \Pi_W$ and for all $Y \subset X$, $\Pi_Y \neq \Pi_W$. It is interesting to note that there may exist more than one minimum partitioner sets of W , as we illustrate in the following example.

Example 4.2. In the 2-dimensional vector space over $\mathbb{Z}_3$, if $W = \{(1,0), (2,0)\}$ then $Min(W) = \{(1,0)\}$ and $Min(W) = \{(2,0)\}$.

In the following result, we provide some properties of maximum and minimum partitioners and show how the partial order between partitions induced by two sets is determined based on their corresponding maximum partitioner sets. Moreover, we provide relation between dispensable set of W with its associated maximum and minimum partitioner sets.

Proposition 4.6. For $W \subseteq \mathcal{V}$, the following statements hold.

1. If $\{W_j \in [W]_{\approx} : j \in \{1, 2, \dots, l\}\}$ then $\cup W_j \in [W]_{\approx}$.
2. If $\Pi_W = \Pi_{\mathcal{V}}$ then $Max(W) = \mathcal{V}$ and $|Min(W) \cap C_b| = 1$ for all $b \in B$.
3. If $\Pi_W \neq \Pi_{\mathcal{V}}$ then $Max(W) = \cup_{w \in W} C_w \cup C_v \cup T_n \cup \{\theta\}$, where $S_v = \cup_{w \in W} S_w$, and $|Min(W) \cap C_w| = 1$ for all $w \in W \setminus \{\theta\}$.
4. $W \subseteq Max(W)$ and $Min(W) \subseteq W$.
5. $Max(W) = Min(W) \cup Disp(W)$ and $Min(W) = Max(W) \setminus Disp(W)$.
6. If $\hat{W} \subseteq \mathcal{V}$, then $\Pi_W \leq \Pi_{\hat{W}} \iff Max(\hat{W}) \subseteq Max(W)$.

Proof. (1) Since each $W_j \in [W]_{\approx}$ which implies $\Pi_{W_j} = \Pi_W$, so $\Pi_{\cup W_j} = \Pi_W$ for $\cup W_j$. Hence, $\cup W_j \in [W]_{\approx}$.

(2) Proof follows from Proposition 3.4 and Theorem 3.3.

(3) Let $u \equiv_W \hat{u}$, which implies $\mathcal{F}(u, w) = \mathcal{F}(\hat{u}, w)$ for all $w \in W$. For all $\hat{u} \in C_w$ corresponding to each $w \in W$, $\mathcal{F}(u, \hat{u}) = \mathcal{F}(\hat{u}, \hat{u})$, so for all $x \in \cup_{w \in W} C_w$ $u \equiv_x \hat{u}$. Now let $v \in \mathcal{V}$ such that $S_v = \cup_{w \in W} S_w$, then $\mathcal{F}(u, v) = \mathcal{F}(\hat{u}, v)$. Moreover, $\mathcal{F}(u, y) = \mathcal{F}(\hat{u}, y) = 1$, for all $y \in T_n$ and $\mathcal{F}(u, \theta) = \mathcal{F}(\hat{u}, \theta) = 0$. Hence, $Max(W) = \cup_{w \in W} C_w \cup C_v \cup T_n \cup \{\theta\}$. Let $u \equiv_W \hat{u}$, which implies $\mathcal{F}(u, w) = \mathcal{F}(\hat{u}, w)$ for all $w \in W$. For all $\hat{u} \in C_w$ corresponding to each $w \in W$ $\mathcal{F}(u, \hat{u}) = \mathcal{F}(\hat{u}, \hat{u})$, so $|Min(W) \cap C_w| = 1$ for all $w \in W$. Since $\mathcal{F}(v, \theta) = \mathcal{F}(\hat{u}, \theta) = 0$ for any $v, \hat{u} \in \mathcal{V}$, which implies if $\theta \in W$ then $\Pi_W = \Pi_{W \setminus \{\theta\}}$, so $\theta \notin Min(W)$. Hence $|Min(W) \cap C_w| = 1$ for all $w \in W \setminus \{\theta\}$.

(4) The proof follows from Proposition 4.6 (3).

(5) The proof follows directly from the definitions of $Max(W)$, $Min(W)$ and $Disp(W)$.

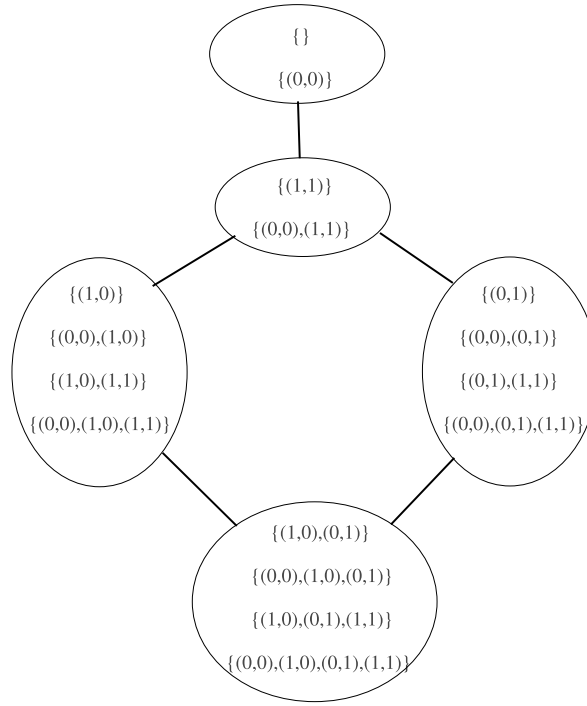
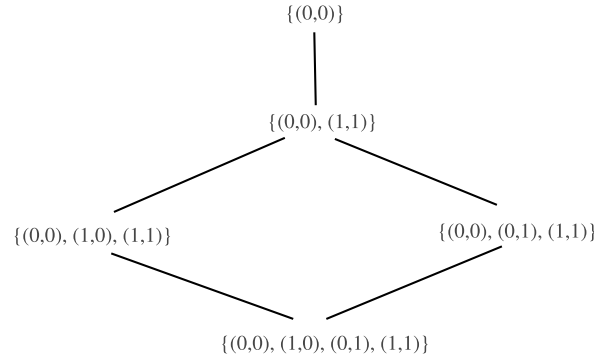
(6) The proof follows from Proposition 4.6 (4) and from equation (5). $\square$

The following result follows from the above result.

Corollary 4.1. For an information system I_B ,

1. $Max(B) = \mathcal{V}$.
2. $Max(W) = \{\theta\}$, where $W = \{\theta\}$.
3. $Max(W) = \{\theta\} \cup T_n$, where $W \subseteq T_n$.

Different lattice structures associated with an information table in terms of indiscernibility relation are discussed in [6]. Analogously, we define two posets for vector space $\mathcal{V}$.

Fig. 2. $\mathcal{L}_{\approx}$.Fig. 3. $\mathcal{L}_{max}$.

Let $\mathcal{L}_{\approx} = (Z, \subseteq^*)$, where $Z = \{[W]_{\approx} : W \subseteq \mathcal{V}\}$ and $\subseteq^*$ is the dual inclusion order, be the poset associated with indiscernible subsets of $\mathcal{V}$ and $\mathcal{L}_{max} = (M(W), \subseteq^*)$, where $M(W) = \{Max(W) : W \subseteq \mathcal{V}\}$, be the poset associated with maximum partitioner sets $Max(W)$.

Example 4.3. For $\mathcal{V}$ over $\mathbb{Z}_2$ with $n = 2$ and $B = \{(1,0), (0,1)\}$, the two lattices $\mathcal{L}_{\approx}$ and $\mathcal{L}_{max}$ are shown in Figs. 2 and 3, respectively.

The following result is a consequence of Proposition 4.6 (4).

Corollary 4.2. For any $X \in M(W)$, $X = Max(X)$.

The above result shows that each set belonging to $M(W)$ is maximum partitioner of itself. Now we show that the union of two maximum partitioner sets is not a maximum partitioner. The following example illustrates that in the case of vector spaces having dimension $n \geq 3$, $M(W)$ is not a union-closed family.

Example 4.4. For $\mathcal{V}$ over $\mathbb{Z}_2$ with $n = 3$, $Max(\{(1,0,0), (0,1,0)\}) = \{\theta, (1,0,0), (0,1,0), (1,1,1)\}$ and $Max(\{(1,0,0), (0,0,1)\}) = \{\theta, (1,0,0), (0,0,1), (1,1,1)\}$, then

$Max(\{(1,0,0), (0,1,0)\}) \cup Max(\{(1,0,0), (0,0,1)\}) = \{\theta, (1,0,0), (0,1,0), (0,0,1), (1,1,1)\}$, which is not a maximum partitioner, because $Max(\{\theta, (1,0,0), (0,1,0), (0,0,1), (1,1,1)\}) = \mathcal{V}$.

Analogous to the definition of the join and meet for maximum partitioner lattice in [6], we have the following definitions in our context.

Definition 4.1. Let $\{W_k : k \in \{1, 2, \dots, l\}\}$ be the collection of all subsets of $\mathcal{V}$, then

1. The join of $\{Max(W_k) : k \in \{1, 2, \dots, l\}\}$ is $\cap_{k=1}^l Max(W_k)$.
2. The meet of $\{Max(W_k) : k \in \{1, 2, \dots, l\}\}$ is $Max(\cup_{k=1}^l Max(W_k))$.

In the following result, we prove that $\mathcal{L}_{max}$ is a complete lattice.

Theorem 4.1. The poset $\mathcal{L}_{max}$ is a complete lattice.

Proof. Let $\{W_k : k \in \{1, 2, \dots, l\}\}$ be collection of all subsets of $\mathcal{V}$ and $X = \{Max(W_k) : k \in \{1, 2, \dots, l\}\}$ be the collection of all maximum partitioners of $\mathcal{V}$. To prove that $\mathcal{L}_{max}$ is a complete lattice, we have to show that the lattice $\mathcal{L}_{max}$ is a join and meet-semi lattice.

First we prove that $\mathcal{L}_{max}$ is a join semi-lattice, for this we first by prove that $\cap_{k=1}^l Max(W_k) \in X$ for this we need to show that $Max(\cap_{k=1}^l Max(W_k)) = \cap_{k=1}^l Max(W_k)$. From Proposition 4.6 (4), $\cap_{k=1}^l Max(W_k) \subseteq Max(\cap_{k=1}^l Max(W_k))$, now it is left to prove the reverse inclusion only. Suppose contrary that there exist $w \in Max(\cap_{k=1}^l Max(W_k))$ such that $w \notin \cap_{k=1}^l Max(W_k)$, which implies $w \notin Max(W_k)$ for some $k \in \{1, 2, \dots, l\}$, and from Proposition 4.6 (3), $w \notin W_k$. Let there exist $v, \hat{v} \in \mathcal{V}$ such that $v \equiv_{Max(W_k)} \hat{v}$, then $v \not\equiv_w \hat{v}$ because $w \notin Max(W_k)$. Now, from $v \equiv_{Max(W_k)} \hat{v}$ we deduce that $v \equiv_{\cap_{k=1}^l Max(W_k)} \hat{v}$, which implies $v \equiv_w \hat{v}$ and contradicts our assumption. Hence, $Max(\cap_{k=1}^l Max(W_k)) = \cap_{k=1}^l Max(W_k)$. Now, as $\cap_{k=1}^l Max(W_k) \subseteq Max(W_k)$ for all $k \in \{1, 2, \dots, l\}$, which implies that $\cap_{k=1}^l Max(W_k)$ is the least upper bound in X . Therefore, the join of all $Max(W_k) \in X$ exists in X , which proves that $\mathcal{L}_{max}$ is a join semi-lattice.

Now, to prove that $\mathcal{L}_{max}$ is a meet semi-lattice, we have to show that $Max(\cup_{k=1}^l Max(W_k)) \in X$. As $Max(W_k) \subseteq \cup_{k=1}^l Max(W_k)$ for all $k \in \{1, 2, \dots, l\}$, which implies $Max(Max(W_k)) \subseteq Max(\cup_{k=1}^l Max(W_k))$ is the greatest lower bound in X . Now, suppose $Y \in X$ such that $Max(\cup_{k=1}^l Max(W_k)) \subseteq Y$, then from equation (4) and Proposition 4.6 (6), $Max(\cup_{k=1}^l Max(W_k)) \subseteq Max(Y) = Y$, which implies that $Max(\cup_{k=1}^l Max(W_k))$ is meet of X . $\square$

Isomorphism between lattices provides information about the structural similarities amongst lattices. Although, the structure of two isomorphic lattices is same but the difference lies in the lattice elements. In the following results, we prove that the three lattices $\mathcal{L}_{\approx}$, $\mathcal{L}_{max}$, and $\mathcal{L}_{\Pi}$ are isomorphic lattices.

Proposition 4.7. The two posets $\mathcal{L}_{\approx}$ and $\mathcal{L}_{max}$ are isomorphic.

Proof. Let $\eta : Z \rightarrow M(W)$ be defined as: $\eta([W]_{\approx}) = Max(W)$. Since for each subset of $\mathcal{V}$ has a unique maximum partitioner, so this implies η is a bijective map. Now we have to prove that η is an isomorphism. Let $W_1, W_2 \subseteq \mathcal{V}$ such that $\Pi_{W_1} = \Pi_{W_2}$ which implies $[W_1]_{\approx} = [W_2]_{\approx}$. Since each $X \in Z$ corresponds to a partition, and for each partition there exist only one maximum partitioner, so this implies $Max(W_1) = Max(W_2)$. Therefore, η is an isomorphism, hence $\mathcal{L}_{\approx}$ and $\mathcal{L}_{max}$ are isomorphic. $\square$

Proposition 4.8. The two posets $\mathcal{L}_{\Pi}$ and $\mathcal{L}_{max}$ are isomorphic.

Proof. Let $\zeta : \Lambda_W \rightarrow M(W)$ be defined as: $\zeta(\Pi_W) = Max(W)$. Since for each Π_W , the corresponding set W has a unique $Max(W)$ which implies ζ is a bijective map. Now we have to prove that ζ is an isomorphism. Let $\Pi_{W_1}, \Pi_{W_2} \in \Lambda_W$, then from equation (6) we have $\Pi_{W_1} \leq \Pi_{W_2}$, which implies $W_2 \subseteq W_1$, and from Proposition 4.6 (6), we have $Max(W_2) \subseteq Max(W_1)$. Now, to complete the proof we have to prove that $Max(W_2) \subseteq Max(W_1)$ implies $\Pi_{W_1} \leq \Pi_{W_2}$, which follows directly from Proposition 4.6 (6). Therefore, ζ is an isomorphism, and hence the two posets $\mathcal{L}_{\Pi}$ and $\mathcal{L}_{max}$ are isomorphic. $\square$

Proposition 4.9. The two posets $\mathcal{L}_{\Pi}$ and $\mathcal{L}_{\approx}$ are isomorphic.

Proof. From Proposition 4.7 and 4.8, we have that the posets $\mathcal{L}_{\Pi}$ and $\mathcal{L}_{\approx}$ are isomorphic to $\mathcal{L}_{max}$, which implies that the two posets $\mathcal{L}_{\Pi}$ and $\mathcal{L}_{\approx}$ are isomorphic. $\square$

The isomorphic property among these lattices provides information about the relationship between their corresponding lattice elements, which are at the same hierarchical level. For instance, in Fig. 3, the topmost lattice element is the maximum partitioner corresponding to the topmost partition in Fig. 1.

Now, we prove that the posets $\mathcal{L}_{\approx}$ and $\mathcal{L}_{\Pi}$ are complete lattices.

Proposition 4.10. *The two posets $\mathcal{L}_{\approx}$ and $\mathcal{L}_{\Pi}$ are complete lattices.*

Proof. From Propositions 4.7, 4.8 and 4.9 it is clear that the three posets $\mathcal{L}_{max}$, $\mathcal{L}_{\approx}$ and $\mathcal{L}_{\Pi}$ are isomorphic and from Theorem 4.1, the poset $\mathcal{L}_{max}$ is a complete lattice. Therefore, from the isomorphism of posets it follows that $\mathcal{L}_{\approx}$ and $\mathcal{L}_{\Pi}$ are also complete lattices. $\square$

The completeness of these lattices shows that the structure of the lattices is well-defined, such that each of these lattices has a minimum and a maximum element within it.

5. Approximations, rough membership function and accuracy measure

In this section, we examine lower and upper approximations of different subsets of $\mathcal{V}$. The lower approximation of a set consists of those objects within the set which certainly belong to the set based on a given indiscernibility relation. The upper approximation consist of those objects which does not belong with certainty in the set based on a given indiscernibility relation. Moreover, we define accuracy of approximation of subsets of $\mathcal{V}$ which quantifies ratio of the number of vectors which are classified with certainty in the set. Furthermore, we define rough membership function of vectors to quantify degree of a vector in a subset of $\mathcal{V}$.

We define the lower and upper approximation of a subset $X \subseteq \mathcal{V}$.

Definition 5.1. For a basis B and $W \subseteq \mathcal{V}$, the lower and upper approximation of $Y \subseteq \mathcal{V}$ is defined respectively as

$$L_{W|B}(Y) = \{u \in \mathcal{V} : [u]_{W|B} \subseteq Y\},$$

$$U_{W|B}(Y) = \{u \in \mathcal{V} : [u]_{W|B} \cap Y \neq \emptyset\}.$$

We say $Y \subseteq \mathcal{V}$ is W_B -exact, if $L_{W|B}(Y) = U_{W|B}(Y)$; otherwise, it is W_B -rough. If the basis is understood from the context, we denote lower and upper approximation by $L_W(Y)$ and $U_W(Y)$ respectively, and write Y is W -rough or W -exact.

Clearly, $L_W(Y) \subseteq Y \subseteq U_W(Y)$, for any set $Y \subseteq \mathcal{V}$, which provides the sufficient condition to check exactness of a subset $Y \subseteq \mathcal{V}$, such that Y is W -exact if $L_W(Y) = Y$; otherwise, it is W -rough. Note that for any basis, $\mathcal{V}$ is W -exact for any $W \subseteq \mathcal{V}$. Moreover, for any two distinct basis $B, \hat{B}$ of $\mathcal{V}$, if $\Pi_{\mathcal{V}|B} = \Pi_{\mathcal{V}|\hat{B}}$, then for any $Y, W \subseteq \mathcal{V}$ $L_{W|B}(Y) = L_{W|\hat{B}}(Y)$ and $U_{W|B}(Y) = U_{W|\hat{B}}(Y)$.

In the following result, we provide the condition under which a subset Y of $\mathcal{V}$ is $\mathcal{V}$ -exact.

Proposition 5.1. *A subset $Y \subseteq \mathcal{V}$ is $\mathcal{V}$ -exact if $Y = \cup_{v \in \mathcal{V}} C_v$ for any arbitrary number of vectors $v \in \mathcal{V}$.*

Proof. Since each block of $\Pi_{\mathcal{V}}$ consists of vectors having the same skeleton. So, if $Y = \cup_{v \in \mathcal{V}} C_v$ for any arbitrary number of vectors $v \in \mathcal{V}$, then $L_W(Y) = Y$. Hence, Y is $\mathcal{V}$ -exact. $\square$

The following remark illustrates the condition under which a subset of $\mathcal{V}$ consisting of only null vector is exact.

Remark 5.1. For $W \subseteq \mathcal{V}$, if $\cup_{w \in W} S_w = B$ then $Y = \{\emptyset\}$ is W -exact, because there does exist not exist $v \in \mathcal{V} \setminus \{\emptyset\}$, such that $\mathcal{F}(v, u) = 0$ for all $u \in \mathcal{V} \setminus \{\emptyset\}$.

The following remark shows which subsets of $\mathcal{V}$ are exact or rough with respect to $W = \{\emptyset\}$.

Remark 5.2. Since all the vectors are indiscernible with respect to $W = \{\emptyset\}$, which implies only $\mathcal{V}$ is W -exact. Moreover, any $Y \subset \mathcal{V}$ is W -rough because $[v]_W = \mathcal{V}$ for any $v \in \mathcal{V}$, which implies $L_W(Y) = \emptyset$ and $U_W(Y) = \mathcal{V}$.

In the following result, we discuss different subsets of $\mathcal{V}$, which are W_B -exact when $B \subseteq W$.

Theorem 5.1. *For I_B and $Y \subseteq \mathcal{V}$, Y is W_B -exact if $B \subseteq W$ and*

$$Y = \begin{cases} T_i, & i \in \{1, 2, \dots, n\} \\ \cup_{v \in \mathcal{V}} C_v, & v \in \mathcal{V} \\ \{\emptyset\}. \end{cases}$$

Proof. Suppose $W \subseteq \mathcal{V}$ such that $B \subseteq W$ then from Proposition 3.4, $\Pi_W = \Pi_{\mathcal{V}}$, and we know that each indiscernibility block of $\Pi_{\mathcal{V}}$ consists of vectors of same skeleton.

- (1) Let $Y = T_i$ for some $i \in \{1, 2, \dots, n\}$, since $\Pi_W = \Pi_{\mathcal{V}}$ which implies $[v]_W \subseteq T_i$ for each $v \in T_i$. Therefore, $L_W(Y) = Y$, which implies Y is W_B -exact.
- (2) Let $\cup_{v \in \mathcal{V}} C_v$ for some arbitrary number of vectors v , since $\Pi_W = \Pi_{\mathcal{V}}$ which implies $[v]_W = C_v$ for each $v \in \mathcal{V}$. Therefore, $L_W(Y) = Y$ which implies that Y is W_B -exact.
- (3) The proof follows from Remark 5.1. $\square$

The following result shows the condition on W under which any subset of $\mathcal{V}$ is exact when the underlying field of $\mathcal{V}$ is of cardinality two.

Corollary 5.1. *For an information system I_B associated with $\mathcal{V}$ over F with $|F| = 2$, if $B \subseteq W$ then any subset $Y \subseteq \mathcal{V}$ is W_B -exact.*

In the following result, we provide approximations of a basis B of $\mathcal{V}$ with respect to the complement of B in $\mathcal{V}$.

Theorem 5.2. *For I_B , if $Y = B$ and $W = \mathcal{V} \setminus B$, then:*

1. *If $n = 1$ and $|F| = 2$, then $L_W(Y) = \emptyset$ and $U_W(Y) = \mathcal{V}$.*
2. *If $n = 2$ and $|F| = 2$, then $L_W(Y) = \emptyset$ and $U_W(Y) = \mathcal{V} \setminus \{\theta\}$.*
3. *If $n \geq 2$ and $|F| = 2$, then Y is W_B -exact.*
4. *If $|F| \geq 3$, then $L_W(Y) = \emptyset$ and $U_W(Y) = T_1$.*

Proof. (1) Let $n = 1$, $|F| = 2$, and $Y = B \subset \mathcal{V}$, then from Theorem 3.2, all the vectors are indiscernible with respect to $W = \mathcal{V} \setminus B$, which implies $L_W(Y) = \emptyset$ and $U_W(Y) = \mathcal{V}$.

(2) Let $n = 2$, $|F| = 2$, and $Y = B$. As $W = \mathcal{V} \setminus B$, then from Theorem 3.2, all the vectors $v \in \mathcal{V} \setminus \{\theta\}$ are indiscernible, which implies there exist no vector $u \in \mathcal{V}$ for which $[u]_W \subseteq Y$, so $L_W(Y) = \emptyset$. Moreover, $[\theta]_W \cap Y = \emptyset$ which implies $\theta \notin U_W(Y)$. Therefore, $U_W(Y) = \mathcal{V} \setminus \{\theta\}$.

(3) Let $n \geq 2$, $|F| = 2$ and $W = \mathcal{V} \setminus B$, then from Theorem 3.2, $[v]_W = \{v\}$ for each vector $v \in \mathcal{V}$ with $|S_v| \leq n - 2$. Since $Y = B$, then $|S_u| = 1 \leq n - 2$ for all $u \in Y$ which implies $[u]_W = \{u\}$ for each $u \in Y$. Therefore $L_W(Y) = Y$, and hence $Y = B$ is W_B -exact.

(4) Let $|F| \geq 3$ and $W = \mathcal{V} \setminus B$, then from Theorem 3.2 $\Pi_W = \Pi_{\mathcal{V}}$. Since each block of $\Pi_{\mathcal{V}}$ consists of vectors with the same skeleton, so $[b]_W = \{kb : k \in F \setminus \{0\}\}$ for each $b \in B$, which implies $B \subset \cup_{b \in B} [b]_W$. Hence $L_W(Y) = \emptyset$ and $U_W(Y) = \cup_{b \in B} [b]_W = T_1$. $\square$

In the following result, we describe an exact set and a rough set with respect to T_i , where $i \in \{1, 2, \dots, n\}$.

Proposition 5.2. *If $W = T_i$, where $i \in \{1, 2, \dots, n\}$, then $Y = \cup T_l$ is W -exact and $Y = T_l$ is W -rough, where $n - i + 1 \leq l \leq n$.*

Proof. Suppose $W = T_i$ and $Y = \cup T_l$, where $i \in \{1, 2, \dots, n\}$ and $n - i + 1 \leq l \leq n$, then from Proposition 3.5, all the vectors in $\cup T_l$ are indiscernible. Therefore, $L_W(Y) = Y$ which implies Y is W -exact.

Now, let $Y = T_l$, where $n - i + 1 \leq l \leq n$, then from Proposition 3.5, $[u]_W = \cup_{m=n-i+1}^n T_m$ for all $u \in T_l$. Since $T_l \subset \cup_{m=n-i+1}^n T_m$ which implies $L_W(Y) = \emptyset$, and hence Y is W -rough. $\square$

It is interesting to note that for $W \subseteq T_n$, $\mathcal{V}$ and $\{\theta\}$ are W -exact, but if $Y \subset \mathcal{V} \setminus \{\theta\}$, then Y is W -rough because $L_W(Y) = \emptyset$ and $U_W(Y) = \mathcal{V} \setminus \{\theta\}$.

For $Y, W \subseteq \mathcal{V}$, we define the accuracy function of approximation of Y with respect to W as $\alpha_W(Y) : Y \longrightarrow [0, 1]$, where $\alpha_W(Y) = |L_W(Y)| / |U_W(Y)|$. Approximation accuracy is the degree of ability of W to classify the vectors in Y as indiscernible. Note that if $\alpha_W(Y) = 1$ then this means W classifies all the vectors in Y as indiscernible and implies Y is W -exact; whereas, if $\alpha_W(Y) < 1$ then Y is W -rough.

Some properties of accuracy function in the context of approximations are as follows.

1. If $L_W(Y) = Y$ then $\alpha_W(Y) = 1$.
2. If $L_W(Y) = \emptyset$ then $\alpha_W(Y) = 0$.
3. If $L_W(Y) \subset Y$ then $0 \leq \alpha_W(Y) < 1$.
4. If $Y \neq \emptyset$ and $L_W(Y) \cap U_W(Y) = \emptyset$ then $\alpha_W(Y) = 0$.

Now, we define the rough membership function of a vector $v \in \mathcal{V}$ with respect to $Y, W \subseteq \mathcal{V}$ as

$$\mu_W^Y(v) = \frac{|\{u \in Y : [u]_W = [v]_W\}|}{|\{\tilde{v} \in \mathcal{V} : [\tilde{v}]_W = [v]_W\}|}.$$

Note that for any vector $v \in \mathcal{V}$, $\mu_W^Y(v) \in [0, 1]$. Some properties of the rough membership function of vectors in relation with approximations are as follows.

1. If $v \notin U_W(Y)$ then $\mu_W^Y(v) = 0$.
2. If $L_W(Y) = \emptyset$ then $\mu_W^Y(v) = 0$, for all $v \in Y$.
3. If $v \in Y \setminus L_W(Y)$ then $0 \leq \mu_W^Y(v) < 1$.
4. If $v \in U_W(Y) \setminus L_W(Y)$ then $0 < \mu_W^Y(v) < 1$.
5. If Y is W -exact, then $\mu_W^Y(v) = 1$, for all $v \in Y$.

6. Discernibility matrix and essential sets

In this section, we define the discernibility matrix associated with I which provide information about the discerning features between feature vectors. Moreover, we associated a poset with the set of all distinct entries of discernibility matrix. Further, we identify the extended cores of I , also known as essential sets.

For an information system I_B , we denote the entry in the discernibility matrix $D_{M|B}$ corresponding to vectors u and v by $M_B(u, v)$; whereas, if the basis is understood from the context, we denote the discernibility matrix by D_M and the entries by $M(u, v)$.

We define $\mathcal{H}(u) = \{v \in \mathcal{V} : \mathcal{F}(u, v) = 1\}$. In the next result, we describe the entries of D_M .

Theorem 6.1. *For I associated with $\mathcal{V}$, the entries in D_M are of the form*

$$M(u, v) = \begin{cases} \mathcal{H}(u) \Delta \mathcal{H}(v), & \text{if } S_u \neq S_v, \\ \emptyset, & \text{if } S_u = S_v, \end{cases}$$

where $\mathcal{H}(u) \Delta \mathcal{H}(v) = \{\mathcal{H}(u) \cup \mathcal{H}(v)\} \setminus \{\mathcal{H}(u) \cap \mathcal{H}(v)\}$.

Proof. (1) Let $S_u \neq S_v$ and $w \in \mathcal{V}$ such that $\mathcal{F}(u, w) \neq \mathcal{F}(v, w)$, which implies either $w \in \mathcal{H}(u)$ or $w \in \mathcal{H}(v)$, so $w \in \mathcal{H}(u) \Delta \mathcal{H}(v)$. Therefore, $w \in M(u, v)$ for all $w \in \mathcal{H}(u) \Delta \mathcal{H}(v)$. To complete the proof, we have to show that there exists no vector $\hat{u} \in \mathcal{V} \setminus (\mathcal{H}(u) \Delta \mathcal{H}(v))$ such that $\hat{u} \in M(u, v)$. Suppose contrary that there exists a vector $\hat{u} \in \mathcal{V} \setminus (\mathcal{H}(u) \Delta \mathcal{H}(v))$ such that $\hat{u} \in M(u, v)$. Since $\hat{u} \in \mathcal{V} \setminus \mathcal{H}(u) \Delta \mathcal{H}(v)$, which implies that $\mathcal{F}(u, \hat{u}) = \mathcal{F}(v, \hat{u})$, and contradicts the hypothesis that $\hat{u} \in M(u, v)$ and hence concludes the proof.

(2) If $S_u = S_v$ then $\mathcal{F}(u, w) = \mathcal{F}(v, w)$ for all $w \in \mathcal{V}$, which implies that $M(u, v) = \emptyset$. $\square$

In one-dimensional vector spaces, all the vectors have same skeleton therefore from Theorem 6.1, we have the following result.

Corollary 6.1. *For an information system I associated with one-dimensional vector space, all entries in the discernibility are empty sets.*

It is worth noting that if F is finite then $O(D_M)$ (order of D_M) is finite, such that if $|F| = q$ then $O(D_M) = q^n \times q^n$, whereas if F is infinite, then the corresponding vector space is infinite, and so $O(D_M)$ is infinite. For such infinite vector spaces $\mathcal{V}$, we define a class-based discernibility matrix $\mathfrak{D}_M$ corresponding to class-based information system $\bar{I}$ and denote in $\mathfrak{D}_M$ corresponding to vectors $u, v \in \mathcal{V}$ by $M(C_u, C_v)$. The same $\mathfrak{D}_M$ can be defined for I associated with vector spaces over finite fields. Note that $O(\mathfrak{D}_M) = 2^n$, where n is dimension of vector space. Now as the order of $\mathfrak{D}_M$ depends on the dimension n , the order of discernibility matrices is the same for two distinct vector spaces having the same dimension.

A numeric discernibility matrix [7] can be associated with an information system, which provides information about the number of features that discern a pair of objects.

Definition 6.1. We call N_{DM} the numeric discernibility matrix of I , where entries in N_{DM} are defined as $N(u, v) = |M(u, v)|$.

Example 6.1. For $\mathcal{V}$ over $\mathbb{Z}_2$ with $n = 2$ and $B = \{(1, 0), (0, 1)\}$, the matrices D_M and N_{DM} are given below respectively. The discernibility matrices are symmetric, so we write only upper triangular entries only.

$$\begin{pmatrix} \emptyset & \emptyset & \emptyset & \emptyset \\ \emptyset & \{(1, 0), (0, 1)\} & \{(0, 1)\} & \\ & \emptyset & \{(1, 0)\} & \\ & & \emptyset & \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & \\ & 0 & 1 & \\ & & 0 & \end{pmatrix}$$

In the following result, we describe the minimum and maximum entry in the numeric discernibility matrix.

Proposition 6.1. *For I associated with $\mathcal{V}$ over a finite field F with $|F| = q$, the minimum non-zero numeric entry in N_{DM} is $q - 1$, and the maximum numeric entry is $2[(q - 1) + (q - 1) \sum_{m=1}^{n-2} \binom{n-2}{m} (q - 1)^m]$.*

Proof. Suppose $u, \hat{u} \in \mathcal{V}$ such that $S_u \cap S_{\hat{u}} \neq \emptyset$ and $v, \hat{v} \in \mathcal{V}$ such that $S_v \cap S_{\hat{v}} = \emptyset$, which implies $|\mathcal{H}(u) \cap \mathcal{H}(\hat{u})| < |\mathcal{H}(v) \cap \mathcal{H}(\hat{v})|$. Now, suppose $w, \hat{w} \in \mathcal{V}$ such that $S_w \cap S_{\hat{w}} = \emptyset$ and $|B \setminus (S_v \cup S_{\hat{v}})| < |B \setminus (S_w \cup S_{\hat{w}})|$, which implies $|\mathcal{H}(w) \cap \mathcal{H}(\hat{w})| < |\mathcal{H}(v) \cap \mathcal{H}(\hat{v})|$, then from Theorem 6.1, $|M(v, \hat{v})| < |M(w, \hat{w})|$. This implies that the minimum entry in N_{DM} corresponds to vectors having $n - 1$ common basis vectors in their skeletons, whereas the maximum numeric entry corresponds to vectors having skeletons of cardinality one and no common basis vectors in their skeletons.

Now, we prove that $q - 1$ is the minimum entry in $\mathcal{N}_{DM}$. Suppose $s, \hat{s} \in \mathcal{V}$ such that $S_s = B$ and $S_{\hat{s}} = B \setminus \{b\}$ for some $b \in B$, which implies from Theorem 6.1, that $\mathcal{M}(s, \hat{s}) = \{w \in \mathcal{V} : S_w = \{b\}\}$. Since $|F| = q$, so $|\mathcal{M}(s, \hat{s})| = q - 1$, therefore $q - 1$ is the minimum non-zero entry in $\mathcal{N}_{DM}$.

Now, we prove that $2[(q - 1) + (q - 1) \sum_{m=1}^{n-2} \binom{n-2}{m} (q - 1)^m]$ is the maximum entry in $\mathcal{N}_{DM}$. Suppose $x, \hat{x} \in \mathcal{V}$ such that $S_x = \{b_i\}$ and $S_{\hat{x}} = \{b_j\}$, where $i \neq j$. It is obvious that all those vectors $t \in \mathcal{V}$ such that $S_t = \{b_i, b_j\}$, does not belong to $\mathcal{M}(x, \hat{x})$. Now, since $|F| = q$ which implies that there exist $(q - 1) \sum_{m=1}^{n-2} \binom{n-2}{m} (q - 1)^m$ vectors $y \in \mathcal{H}(x)$ such that $|S_y| \geq 2$ and $b_j \notin S_y$, whereas there exist $q - 1$ vectors in $\mathcal{H}(x)$ having skeleton $\{b_i\}$. Therefore, there exist $(q - 1) + (q - 1) \sum_{m=1}^{n-2} \binom{n-2}{m} (q - 1)^m$ vectors z in $\mathcal{H}(x)$ such that $b_j \notin S_z$. Similarly, there exist $(q - 1) + (q - 1) \sum_{m=1}^{n-2} \binom{n-2}{m} (q - 1)^m$ vectors $\hat{z}$ in $\mathcal{H}(x)$ such that $b_i \notin S_{\hat{z}}$. Hence, the maximum numeric entry is $2[(q - 1) + (q - 1) \sum_{m=1}^{n-2} \binom{n-2}{m} (q - 1)^m]$. $\square$

The following results follow directly from Proposition 6.1.

Corollary 6.2. For I associated with vector space of dimension one, $\mathcal{N}_{DM}$ is a null matrix.

Corollary 6.3. For I associated with a vector space over an infinite field, there is no finite numeric entry in $\mathcal{N}_{DM}$ other than zero.

The discernibility function associated with D_M is defined as $f_{dis} = \bigwedge \bigvee \mathcal{M}(u, v)$. In [26], Pawlak proved that if entries of a discernibility matrix contain singleton sets then the core coincides with the collection of all singleton entries of the discernibility matrix.

Proposition 6.2. For I_B associated with $\mathcal{V}$ over field F with $|F| = 2$, $\text{Core}(I_B) = B$ and $|\text{Core}(I_B)| = n$.

Proof. Since $|F| = 2$ so all the vectors have different skeletons, which implies $|C_u| = 1, \forall u \in \mathcal{V}$. Let $v, \hat{v} \in \mathcal{V}$ such that $|S_v \cap S_{\hat{v}}| = n - 1$, then from Theorem 6.1, $\mathcal{M}(v, \hat{v}) = \{b_k\}$ such that $b_k \in S_v$ or $b_k \in S_{\hat{v}}$, where $k \in \{1, 2, \dots, n\}$. Hence for all $i \in \{1, 2, \dots, n\}$, $\{b_i\}$ are the singleton entries in D_M , which implies $\text{Core}(I_B) = \bigcup_{i=1}^n \{b_i\} = B$, and $|\text{Core}(I_B)| = n$. $\square$

Note that the core of information systems associated with $\mathcal{V}$ over F with $|F| \geq 3$ is an empty set, as we illustrate in the following example.

Example 6.2. Let $\mathcal{V}$ be a vector space of dimension two over $\mathbb{Z}_3$ and $B = \{(1, 0), (0, 1)\}$.

$\text{DISC}(I) = \{\{\}, C_{(1,0)}, C_{(0,1)}, C_{(1,0)} \cup C_{(0,1)}\} = \{\{\}, \{(1, 0), (2, 0)\}, \{(0, 1), (0, 2)\}, \{(1, 0), (2, 0), (0, 1), (0, 2)\}\}$. From discernibility function the reducts are computed as:

$f_{dis} = \{(1, 0) \vee (2, 0)\} \wedge \{(0, 1) \vee (0, 2)\} \wedge \{(1, 0) \vee (2, 0) \vee (0, 1) \vee (0, 2)\}$.

As, $C_{(1,0)} \subseteq \{C_{(1,0)} \cup C_{(0,1)}\}$ and $C_{(0,1)} \subseteq \{C_{(1,0)} \cup C_{(0,1)}\}$, so f_{dis} is simplified to:

$f_{dis} = \{(1, 0) \vee (2, 0)\} \wedge \{(0, 1) \vee (0, 2)\}$.

After applying conjunction $\vee$ and disjunction $\wedge$, we obtain four reducts as:

$\{(1, 0), (0, 1)\}, \{(1, 0), (0, 2)\}, \{(2, 0), (0, 1)\}, \{(2, 0), (0, 2)\}$.

Moreover, the core is an empty set. The class-based discernibility matrix $\mathfrak{D}_M$ is given below, since discernibility matrices are symmetric so we write only upper triangular entries only.

$$\begin{pmatrix} \emptyset & \emptyset & \emptyset & \emptyset \\ & \emptyset & C_{(1,0)} \cup C_{(0,1)} & C_{(0,1)} \\ & & \emptyset & C_{(1,0)} \\ & & & \emptyset \end{pmatrix}$$

With every information system, a discernibility poset [7] can be associated, where the elements of this poset consist of all the distinct entries from the associated discernibility matrix. We define the poset associated with the collection of all distinct entries $\text{DISC}(I)$ of D_M as: $\mathfrak{D} = (\mathcal{L}, \subseteq)$, where $\mathcal{L} = \{W \subseteq \mathcal{V} \mid W \in \text{DISC}(I)\}$. For $U_1, U_2 \in \mathcal{L}$, we define the join and meet in $\mathfrak{D}$ respectively as: $U_1 \wedge U_2 = U_1 \cup U_2$ and $U_1 \vee U_2 = U_1 \cap U_2$.

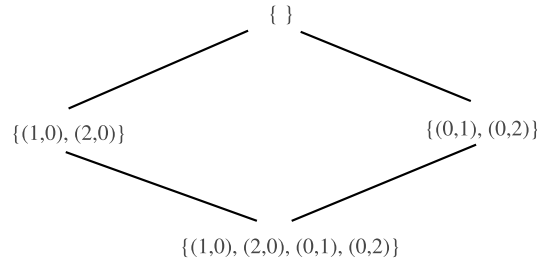
The following result describes the condition for which $\mathfrak{D}$ is a complete lattice.

Proposition 6.3. For I associated with $\mathcal{V}$ over F with $n = 2$ and $|F| \geq 2$, $\mathfrak{D}$ is a complete lattice.

Proof. Let $B = \{b_1, b_2\}$, and $u, v, w \in \mathcal{V}$ such that $S_u = \{b_1\}$, $S_v = \{b_2\}$ and $S_w = B$, which implies $\mathcal{M}(C_u, C_v) = C_{b_1} \cup C_{b_2} = T_1$, $\mathcal{M}(C_u, C_w) = C_{b_2}$ and $\mathcal{M}(C_v, C_w) = C_{b_1}$, so $\text{DISC}(I) = \{\emptyset, C_{b_1}, C_{b_2}, T_1\}$. Hence $U_1 \wedge U_2 \in \text{DISC}(I)$ and $U_1 \vee U_2 \in \text{DISC}(I)$ for any $U_1, U_2 \in \text{DISC}(I)$, which implies $\mathfrak{D}$ is complete. $\square$

The discernibility lattice of $\mathcal{V}$ over $\mathbb{Z}_3$ with $n = 2$ is shown in Fig. 4.

Note that for a vector space with $n \geq 3$, $\mathfrak{D}$ is not complete as we illustrate in the following example.

Fig. 4. $\mathfrak{L}_D$.

Example 6.3. Let $\mathcal{V}$ over $\mathbb{Z}_2$ with $n=3$ and $B = \{(1,0,0), (0,1,0), (0,0,1)\}$.

For $\mathcal{M}((1,0,0), (1,1,0)) = C_{(0,1,0)} \cup C_{(0,1,1)}$ and $\mathcal{M}((1,0,0), (1,0,1)) = C_{(0,0,1)} \cup C_{(0,1,1)}$, the meet is $C_{(0,1,1)} \notin DISC(I)$. Now, for $\mathcal{M}((1,0,0), (0,1,0)) = C_{(1,0,0)} \cup C_{(0,1,0)} \cup C_{(1,0,1)} \cup C_{(0,1,1)}$ and $\mathcal{M}((1,0,0), (0,0,1)) = C_{(1,0,0)} \cup C_{(0,0,1)} \cup C_{(1,1,0)} \cup C_{(0,1,1)}$, the join is $\{C_{(1,0,0)} \cup C_{(0,1,0)} \cup C_{(0,0,1)} \cup C_{(0,1,1)} \cup C_{(1,1,0)} \cup C_{(1,0,1)}\} \notin DISC(I)$.

An essential set [7] is a subset of features with the property that if it is removed from the feature set then the resulting set of features does not preserve the classification of objects obtained with respect to complete set of features. Now, we define the essential sets of the information system I associated with $\mathcal{V}$.

Definition 6.2. A subset $E \subseteq \mathcal{V}$ is called essential of I_B , if $\Pi_{\mathcal{V} \setminus E} \neq \Pi_{\mathcal{V}}$ and $\Pi_{\mathcal{V} \setminus F} = \Pi_{\mathcal{V}}$, $\forall F \subsetneq E$.

We denote by $Ess(I_B)$, the collection of all essential sets associated with I_B , and define the collection of k -essential sets and the essential dimension of I_B respectively as

$$Ess_k(I_B) = \{E \in Ess(I_B) : |E| = k\},$$

$$Ess_{dim}(I_B) = \min\{k : Ess_k(I_B) \neq \emptyset\}.$$

The following result provides the essential sets of I_B associated with $\mathcal{V}$ and shows that the essential dimension depends on $|F|$, whereas the number of essential sets $Ess(I_B)$ depends on the dimension of $\mathcal{V}$.

Theorem 6.2. For an information system I_B , where $B = \{b_1, b_2, \dots, b_n\}$, the following statements hold.

1. $Ess(I_B) = \{B_1, B_2, \dots, B_n\}$
2. $Ess_{dim}(I_B) = |F| - 1$
3. $|Ess(I_B)| = n$.

Proof. (1) Let $B = \{b_1, b_2, \dots, b_n\}$ and for each $b_i \in B$, $C_{b_i} = B_i$. From Theorem 3.4, $\Pi_B = \Pi_{\mathcal{V}}$ which implies for any $i \in \{1, 2, \dots, n\}$, $\Pi_{\mathcal{V} \setminus B_i} \neq \Pi$ because there exist $u \in \mathcal{V}$ with $S_u = B \setminus \{b_i\}$ such that $u \equiv_{\mathcal{V} \setminus B_i} v$, where $v \in T_n$. If $A \subset B_i$ for some $i \in \{1, 2, \dots, n\}$ then from Theorem 3.4, $\Pi_{\mathcal{V} \setminus A} = \Pi_{\mathcal{V}}$. Therefore, each B_i is an essential set. Now we prove that B_i are the only essential sets, suppose contrary that there exists a subset $X \subseteq \mathcal{V}$ which is an essential set of I_B such that $X \neq B_i$ for all $i \in \{1, 2, \dots, n\}$. Since $X \neq B_i$, so we have:

Case 1: Let $X \cap B_i = \emptyset$, $\forall i \in \{1, 2, \dots, n\}$, which implies $X \cap B = \emptyset$ then from Proposition 3.4 we deduce that $\Pi_{\mathcal{V} \setminus X} = \Pi_{\mathcal{V}}$ which contradicts the first condition of the essential set. Hence X is not an essential set.

Case 2: Let $X \subset \cup_{i=1}^n B_i \neq \emptyset$ which implies from Theorem 3.3 $\Pi_{\mathcal{V} \setminus X} = \Pi_{\mathcal{V}}$ which contradicts the first condition of the essential set. Hence X is not an essential set.

Case 3: Let $X \cap B_i \neq \emptyset$ for at least one $i \in \{1, 2, \dots, n\}$ which implies $X \cap [b_i]_{\mathcal{V}} = \emptyset$ then from Theorem 3.3 $\Pi_{\mathcal{V} \setminus X} \neq \Pi_{\mathcal{V}}$ this completes the first condition of the essential set. Now let $X = Y \cup \dot{Y}$ such that $Y = \{v \in X : |S_v| \geq 2\}$ and $\dot{Y} = \{v \in X : |S_v| = 1\}$ then from Theorem 3.3, $\Pi_{\mathcal{V} \setminus Y} \neq \Pi_{\mathcal{V}}$ which contradicts the second condition of the essential set. Hence X is not an essential set.

(2) From Theorem 6.2 (1), it is clear that B_i are the only essential sets of I_B , for all $i \in \{1, 2, \dots, n\}$. Since each B_i consist of $|F| - 1$ elements, so cardinality of each essential set is $|F| - 1$ which implies $Ess_{dim}(I) = |F| - 1$.

(3) Since dimension of $\mathcal{V}$ is n , so $|B| = n$, and for each $b_i \in B$, $C_{b_i} = B_i$. From Theorem 6.2 (1), each B_i is an essential set, so there exist n essential sets of I_B , so $|Ess(I_B)| = n$. $\square$

Example 6.4. For a 2-dimensional vector space over $\mathbb{Z}_3$ and $B = \{(1,0), (0,2)\}$, the essential sets are $\{(1,0), (2,0)\}$ and $\{(0,1), (0,2)\}$.

The following result follows from Theorem 6.2.

Corollary 6.4. If the field is of infinite cardinality, the cardinality of each essential set and essential dimension is infinite.

The Algorithm 2 computes all the essential sets of I associated with $\mathcal{V}$.

Algorithm 2 Essential sets of an information system.

```

Input:  $Ess(I) = \{E_j \subseteq \mathcal{V} : 1 \leq j \leq s = 2^{|\mathcal{V}|}\}$  ▷ Set of all subsets of  $\mathcal{V}$ .
Output:  $Ess(I)$  ▷ Set of all essential sets of information system  $I$  associated with  $\mathcal{V}$ .
Initialize:  $j = 1$ 
for  $j \leq s$  do
  if  $\Pi_{\mathcal{V} \setminus E_j} = \Pi_{\mathcal{V}}$  then
     $Ess(I) = Ess(I) \setminus E_j$ 
     $j = j + 1$ 
  else
     $j = j + 1$ 
  end if
end for
 $Ess(I) = \{A_l : 1 \leq l \leq k\}$  ▷  $Ess(I)$  updated from the above for loop
 $l = 1$ 
for  $l \leq k$  do
   $H_l = \{F_i \subset A_l \mid 1 \leq i \leq r\}$  ▷ Set of all proper subsets of the set  $A_l \in Ess(I)$ 
  if  $\Pi_{\mathcal{V} \setminus F_i} \neq \Pi_{\mathcal{V}}$ , for at least one  $i \in \{1, 2, \dots, r\}$  then
     $Ess(I) = Ess(I) \setminus A_l$ 
     $l = l + 1$ 
  else
     $l = l + 1$ 
  end if
end for
Return  $Ess(I)$ 

```

It is interesting to note that for $\mathcal{V}$ over field F with $|F| = 2$, the union of all essential sets of I_B of $\mathcal{V}$ is equal to B . Moreover, for $\mathcal{V}$ over field F with $|F| \geq 3$, the union of all essential sets of I_B is a superset of the basis B .

In the following result, we discuss the condition on the field for which the core coincides with the collection of all essential sets of I .

Proposition 6.4. For I associated with $\mathcal{V}$ over field F with $|F| = 2$, $Core(I) = \bigcup \{E \mid E \in Ess_1(I)\}$.

Proof. Let $|F| = 2$ and n be dimension of $\mathcal{V}$ then from Theorem 6.2, $Ess_{dim} = 1$ and $Ess_1(I) = \{b_i\}$, $\forall i \in \{1, 2, \dots, n\}$. Since the core is the collection of singleton entries of a discernibility matrix, therefore $Core(I) = \bigcup \{E : E \in Ess_1(I)\}$. $\square$

In the following result, we prove that the essential sets of an information system I coincide with the minimal entries of its discernibility matrix D_M .

Theorem 6.3. The minimal elements in $DISC(I)$ are the essential sets of I .

Proof. Suppose $E \in Ess(I)$ then from the definition of essential set $\Pi_{\mathcal{V} \setminus E} \neq \Pi_{\mathcal{V}}$ and $\Pi_{\mathcal{V} \setminus F} = \Pi_{\mathcal{V}}$, for all $F \subset E$. Suppose $u, \hat{u} \in \mathcal{V}$ with $S_u \neq S_{\hat{u}}$ and $u \equiv_{\mathcal{V} \setminus E} \hat{u}$. Since $\Pi_{\mathcal{V}}$ is the finest partition and each of its blocks consists of vectors having the same skeleton so this implies $u \not\equiv_{\mathcal{V}} \hat{u}$. As, $u \equiv_{\mathcal{V} \setminus E} \hat{u}$ so $\mathcal{V} \setminus E \subseteq \mathcal{V} \setminus M(u, \hat{u})$, which implies $M(u, \hat{u}) \subseteq E$. Now to prove the reverse inclusion, suppose contrary that there exists $e \in E$ such that $e \notin M(u, \hat{u})$ which implies $u \equiv_e \hat{u}$. Suppose $A = E \setminus \{e\}$, as $u \equiv_{\mathcal{V} \setminus E} \hat{u}$ and $u \equiv_e \hat{u}$ which implies $\mathcal{F}(u, x) = \mathcal{F}(\hat{u}, x)$ for all $x \in \mathcal{V} \setminus A$. Now as $u \not\equiv_{\mathcal{V}} \hat{u}$, so we have $\Pi_{\mathcal{V} \setminus A} \neq \Pi_{\mathcal{V}}$ but this contradicts our hypothesis that $E \in Ess(I)$. Therefore, our supposition is wrong, so $e \in M(u, \hat{u})$, and $E \subseteq M(u, \hat{u})$. Hence $E \in DISC(I)$. Now to prove that E is minimal in $DISC(I)$, suppose contrary that there exist $\hat{E} \in DISC(I)$ such that $\hat{E} \subset E$. Now as $\hat{E} \in DISC(I)$ so this implies there exists $y, \hat{y} \in \mathcal{V}$ with $S_y \neq S_{\hat{y}}$, and $E = M(y, \hat{y})$. Since $S_v \neq S_{\hat{v}}$ which implies $v \not\equiv_{\mathcal{V}} \hat{v}$, so $\Pi_{\mathcal{V} \setminus \hat{E}} \neq \Pi_{\mathcal{V}}$ which contradicts our hypothesis that $E \in Ess(I)$. So, our supposition is wrong and this proves that E is minimal in $DISC(I)$ and each $E \in Ess(I)$ is minimal. Now to complete the proof, we need to show that every minimal set in $DISC(I)$ is an essential set. Let D be a minimal set in $DISC(I)$ and there exist $z, \hat{z} \in \mathcal{V}$ such that $D = M(z, \hat{z})$ and $\mathcal{F}(z, d) \neq \mathcal{F}(\hat{z}, d)$ for all $d \in D$. As $D \subset \mathcal{V}$ so this implies $z \not\equiv_{\mathcal{V}} \hat{z}$ and $z \equiv_{\mathcal{V} \setminus D} \hat{z}$, which means $\Pi_{\mathcal{V} \setminus D} \neq \Pi_{\mathcal{V}}$, and so first condition of the essential set is proved. Now we prove that D fulfills the second condition of essential set, suppose contrary that there exist $C \subset D$ such that $\Pi_{\mathcal{V} \setminus C} \neq \Pi_{\mathcal{V}}$ then $z \not\equiv_{\mathcal{V}} \hat{z}$ and $z \equiv_{\mathcal{V} \setminus C} \hat{z}$. Now $\mathcal{F}(z, c) = \mathcal{F}(\hat{z}, c)$ for all $c \in \mathcal{V} \setminus C$ so, $\mathcal{V} \setminus C \subseteq \mathcal{V} \setminus M(z, \hat{z})$ which implies $M(z, \hat{z}) \subseteq C \subset D$ which contradicts the fact that D is minimal in $DISC(I)$. Therefore, $\Pi_{\mathcal{V} \setminus C} = \Pi_{\mathcal{V}}$ and so D is an essential set. Hence, all the minimal sets in $DISC(I)$ are the essential sets. $\square$

By using the concepts of discernibility matrices, it can be identified which pair of feature vectors behave alike or otherwise and which feature vectors are discerning that pair. The entries of discernibility matrix provide information about the ability of a feature vector to classify other feature vectors and the minimal entries in the discernibility matrix, named as essential sets, help identify redundant vectors as well as reducts.

Table 2
Customers data.

	a_1	a_2	a_3	a_4	a_5
c_1	Card	Yes	Yes	Male	In-store
c_2	Card	No	Yes	Female	Online
c_3	Cash	No	Yes	Female	Online
c_4	Cash	Yes	No	Male	In-store
c_5	Card	No	No	Female	In-store
c_6	Cash	Yes	No	Female	Online
c_7	Card	No	No	Female	Online
c_8	Card	No	Yes	Female	In-store

Table 3
Customers data after transformation.

	a_1	a_2	a_3	a_4	a_5
c_1	1	1	1	1	0
c_2	1	0	1	0	1
c_3	0	0	1	0	1
c_4	0	1	0	1	0
c_5	1	0	0	0	0
c_6	0	1	0	0	1
c_7	1	0	0	0	1
c_8	1	0	1	0	0

7. Applications

In this section, we examine two real-life scenarios using the terminology and concepts presented in the previous sections. The first scenario goes in line with $\mathbb{Z}_2^n$ and the second one goes in line with $\mathbb{Z}_3^n$. It is worth noting that the investigation of $\mathbb{Z}_2^n$ sets tone for the analysis of $\mathbb{Z}_p^n$. The only difference lies in the cardinality of classes in the partition. The idea developed provides frameworks for the analysis of feature spaces like $\mathbb{Z}_p^n$ as well as for n dimensional vector spaces over infinite fields.

7.1. Customer data

The world of business is progressing by data driven decisions. Customers are the most important part of any business without customers there is no business. Customer retention is a big challenge for businesses. Customer retention can be increased by analysing customer behaviour and purchase patterns. Customer segmentation is the process of grouping customers possessing similar characteristics. Customer segmentation helps businesses understand customer behaviours and preferences, which can be used to tailor targeted product recommendations and marketing campaigns.

Consider customer data shown in Table 2, the data consists of eight customers and five features which provide information about customer purchase method. The features and associated feature values for each feature are listed below as:

1. a_1 = Payment method (Card or Cash)
2. a_2 = Used discount code (Yes or No)
3. a_3 = Reviewed a product (Yes or No)
4. a_4 = Gender of customer (Male or Female)
5. a_5 = Mode of purchase (In-store or Online)

Since, the components of a vector of a vector space are numerical values, so to apply the proposed concepts, we first convert the feature vectors of each customer into numerical feature vectors. Suppose we are interested in studying behaviour of male customers who used card for payment and bought product online. According to the goal of analysis, we encode (transform) the feature values. For feature a_1 , we encode Card=1 and Cash=0, for a_2 and a_3 , we encode Yes=1 and No=0, for a_4 we encode Male=1 and Female=0, and for a_5 we encode In-stores=0 and Online=1. The new information system where each customer is represented as a numerical feature vector is given below in Table 3.

Note that the customer c_1 has feature value 1 for the features a_1, a_2, a_3, a_4 , whereas c_2 has feature value 1 for the features a_1, a_3, a_5 and similarly for other customers those features can be identified for which they have non-zero feature values. Now from Definition 3.1, we form a Boolean information system where rows and columns represent feature vectors of customers and the values in the Boolean information system are determined by using equation (2).

The discernibility matrix of Table 4 is given below. Since the discernibility matrices are symmetric with diagonal entries as $\emptyset$ so we presented only upper half of the matrix and omitted first column. The row labels are $c_1, c_2, c_3, c_4, c_5, c_6, c_7, c_8$ and the column labels are $c_2, c_3, c_4, c_5, c_6, c_7, c_8$.

Table 4
Boolean information system of customers data.

	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
c_1	1	1	1	1	1	1	1	1
c_2	1	1	1	0	1	1	1	1
c_3	1	1	1	0	0	1	1	1
c_4	1	0	0	1	0	1	0	0
c_5	1	1	0	0	1	0	1	1
c_6	1	1	1	1	0	1	1	0
c_7	1	1	1	0	1	1	1	1
c_8	1	1	1	0	1	0	1	1

Table 5
Asthma patients data.

	a_1	a_2	a_3	a_4
p_1	High	Medium	Low	Yes
p_2	High	Low	Low	No
p_3	Medium	Low	Medium	Yes
p_4	Low	Low	Medium	No
p_5	Low	Low	Low	Yes
p_6	Medium	High	Low	Yes
p_7	High	Low	Medium	Yes
p_8	Low	High	Low	No

$$\left(\begin{array}{cccccc} \{c_4\} & \{c_4, c_5\} & \{c_2, c_3, c_5, c_7, c_8\} & \{c_3, c_4, c_6\} & \{c_5, c_8\} & \{c_4\} & \{c_4, c_6\} \\ \emptyset & \{c_5\} & \{c_2, c_3, c_4, c_5, c_7, c_8\} & \{c_3, c_6\} & \{c_4, c_5, c_8\} & \emptyset & \{c_6\} \\ & \emptyset & \{c_2, c_3, c_4, c_7, c_8\} & \{c_3, c_5, c_6\} & \{c_4, c_8\} & \{c_5\} & \{c_5, c_6\} \\ & & \emptyset & \{c_2, c_4, c_5, c_6, c_7, c_8\} & \{c_2, c_3, c_7\} & \{c_2, c_3, c_4, c_5, c_7, c_8\} & \{c_2, c_3, c_4, c_5, c_6, c_7, c_8\} \\ & & & \emptyset & \{c_3, c_4, c_5, c_6, c_8\} & \{c_3, c_6\} & \{c_3\} \\ & & & & \emptyset & \{c_4, c_5, c_8\} & \{c_4, c_5, c_6, c_8\} \\ & & & & & \emptyset & \{c_6\} \\ & & & & & & \emptyset \end{array} \right)$$

The essential sets of Table 4 are $\{c_3\}$, $\{c_4\}$, $\{c_5\}$, $\{c_6\}$ which are the minimal entries of the discernibility matrix. We compute the reduct using discernibility function, after applying conjunction $\vee$ (absorption property of discernibility function) the discernibility function f_{dis} is simplified to $f_{dis} = \{c_3\} \wedge \{c_4\} \wedge \{c_5\} \wedge \{c_6\}$ and the reduct obtained is $\{c_3, c_4, c_5, c_6\}$. Since singleton entries of a discernibility matrix form core, therefore $\text{core} = \{c_3, c_4, c_5, c_6\}$. The core and the reduct are the same as expressed in Theorem 3.4. The reduct $\{c_3, c_4, c_5, c_6\}$ provides sufficient information to segment customers having similar characteristics. The customers are segmented as: $c_1|c_2, c_7|c_3|c_4|c_5|c_6|c_8$.

In this example, we have used the terminology of vector space $\mathbb{Z}_2^n$. In the following example, we explain that this idea can be extended to other vector spaces.

7.2. Asthma patients data

Asthma, a chronic respiratory disease, is characterised by inflammation and airway narrowing, causes difficulty in breathing and wheezing. Beyond its symptoms, asthma impacts everyday activities, sleep, and general quality of life. Segmenting asthma patients with similar features can help health practitioners develop strategies to manage the disease of patients by identifying their unique triggers.

Consider data of asthma patients shown in Table 5, the data consists of eight patients and four features. The features and associated feature values for each feature are listed below as:

1. a_1 = Respiratory Rate (High, Medium, Low)
2. a_2 = Wheezing Intensity (High, Medium, Low)
3. a_3 = Airway Inflammation (High, Medium, Low)
4. a_4 = Diagnosis (Yes or No)

To apply the proposed terminology and concepts, we first convert the feature vectors of each patient into numerical feature vectors. Since low respiratory rate, low wheezing intensity and low airway inflammation does not contribute much to diagnosis of asthma so, according to this domain knowledge, we encode (transform) the feature values of a_1, a_2, a_3 as: High = 2, Medium = 1 and Low = 0 and for the feature values of a_4 as: Yes = 1 and No = 0. Analogous to Table 3, we can construct a new table where each patient is represented as a numerical feature vector. Now from Definition 3.1, we form a Boolean information system as shown in Table 6, where rows and columns represent feature vectors of the patients and the values in the Boolean information system are determined using equation (2).

Table 6
Boolean information system of asthma patients data.

	p_1	p_2	p_3	p_4	p_5	p_6	p_7	p_8
p_1	1	1	1	0	1	1	1	1
p_2	1	1	1	0	0	1	1	0
p_3	1	1	1	1	1	1	1	0
p_4	0	0	1	1	0	0	1	0
p_5	1	0	1	0	1	1	1	0
p_6	1	1	1	0	1	1	1	1
p_7	1	1	1	1	1	1	1	0
p_8	1	0	0	0	0	1	0	1

The discernibility matrix of Table 6 is given below. Since the discernibility matrices are symmetric with diagonal entries as $\emptyset$ so we write only upper half of the matrix and omit first column of the discernibility matrix. The row labels are $p_1, p_2, p_3, p_4, p_5, p_6, p_7, p_8$ and the column labels are $p_2, p_3, p_4, p_5, p_6, p_7, p_8$.

$$\begin{pmatrix} \{p_5, p_8\} & \{p_4, p_8\} & \{p_1, p_2, p_4, p_5, p_6, p_8\} & \{p_2, p_8\} & \emptyset & \{p_4, p_8\} & \{p_2, p_3, p_5, p_7\} \\ \emptyset & \{p_4, p_5\} & \{p_1, p_2, p_4, p_6\} & \{p_2, p_5\} & \{p_5, p_8\} & \{p_4, p_5\} & \{p_2, p_3, p_7\} \\ & \emptyset & \{p_1, p_2, p_5, p_6\} & \{p_2, p_4\} & \{p_4, p_8\} & \emptyset & \{p_2, p_3, p_4, p_5, p_7, p_8\} \\ & & \emptyset & \{p_1, p_4, p_5, p_6\} & \{p_1, p_2, p_4, p_5, p_6, p_8\} & \{p_1, p_2, p_5, p_6\} & \{p_1, p_3, p_4, p_6, p_7, p_8\} \\ & & & \emptyset & \{p_2, p_8\} & \{p_2, p_4\} & \{p_3, p_5, p_7, p_8\} \\ & & & & \emptyset & \{p_4, p_8\} & \{p_2, p_3, p_5, p_7\} \\ & & & & & \emptyset & \{p_2, p_3, p_4, p_5, p_7, p_8\} \\ & & & & & & \emptyset \end{pmatrix}$$

The essential sets are $\{p_5, p_8\}, \{p_4, p_8\}, \{p_4, p_5\}, \{p_2, p_8\}, \{p_2, p_5\}, \{p_2, p_4\}$. We compute the reduct using discernibility function, after applying conjunction $\vee$ (absorption property of discernibility function) the discernibility function f_{dis} is simplified to $f_{dis} = \{p_5, p_8\} \wedge \{p_4, p_8\} \wedge \{p_4, p_5\} \wedge \{p_2, p_8\} \wedge \{p_2, p_5\} \wedge \{p_2, p_4\}$. The four reducts obtained from discernibility function are: $R_1 = \{p_2, p_4, p_5\}$, $R_2 = \{p_2, p_4, p_8\}$, $R_3 = \{p_4, p_5, p_8\}$, $R_4 = \{p_2, p_5, p_8\}$ and the core is empty. The information provided by all the patients in a reduct provide sufficient information for segmenting similar patients into groups. The segmentation of this patients data is obtained as: $p_1, p_6 | p_3, p_7 | p_2 | p_4 | p_5 | p_8$.

We associate a feature vector with each object in the data, and using the concepts and terminology introduced in this paper, we segmented data into groups. Our approach helps in grouping up objects having similar features, and the same has been exhibited in these two examples.

8. Conclusions

In this paper, we have studied vector spaces, also called feature space in the field of data science and machine learning, in the context of rough set theory. For this purpose, we have defined an information system associated with vector space and an indiscernibility relation on vector space. The defined indiscernibility relation provides a framework to partition data according to specific analytical goal. We have investigated different partitions of a vector space and identified the relation between basis of the vector space and reducts of the information system. Moreover, we have proved that the canonical partitions with respect to any two distinct basis are size-isomorphic. To investigate hierarchical relation between partitions, we have defined three types of posets and proved that these three posets $\mathcal{L}_\Pi, \mathcal{L}_\approx, \mathcal{L}_{max}$ are isomorphic and complete lattices. Further, we have also studied approximations of different subsets of vector space. Discernibility matrices provide information about discerning properties between data instances. We have provided the general notion of the entries of discernibility matrix for the defined information system and identified that the order of discernibility matrix depends on the cardinality of field and the dimension of vector space. We have also proved that the core of an information system associated with a vector space over a field of cardinality three or more is empty. Moreover, we have identified that the poset associated with distinct entries of a discernibility matrix is complete only for vector spaces of dimension two. An essential set preserves the canonical partition of data such that removing any element from an essential set results in loss of information. We have determined essential sets for the defined information system and proved that the essential sets depend on the underlying basis of the information system and the essential dimension is dependent on the cardinality of field. Moreover, we have identified relation between the minimal entries of the discernibility matrix and the essential sets of the information system. Finally, we have demonstrated the use of developed theory and concepts by providing two practical real life examples.

This study of vector spaces in context of RST can be extended for infinite dimension. Moreover, vector spaces can be studied in context of variants of RST such as variable precision rough sets and dominance-based rough sets. RST provides a strict classification of objects, whereas the variable precision rough sets provide flexible approach for studying indiscernibility among objects by setting up a level of precision to determine indiscernibility of objects. The dominance-based rough sets are based on dominance relation which relates an object based on its features as dominating or dominated by other objects. Moreover, it will be interesting to study vector spaces using multigranulation rough sets by defining more than one relation on vector space. Furthermore, fuzzy information system can be introduced for vector spaces and studied in the context of fuzzy indiscernibility relation. In addition to that, vector spaces can also be studied in context of hybrid models such as fuzzy rough sets, rough fuzzy sets as well.

CRediT authorship contribution statement

Abeer Fatima: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Imran Javaid:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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